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On general relativity:

how to physics in a curved spacetime.

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Abstract

In these notes, we will study general relativity.

Part I

Einstein field equations

Chapter 1

Principles of general relativity

In this chapter, we will briefly review all the underline principles that led Einstein to formulate its theory: principle of general relativity, principles of equivalence and principle of general covariance.

1.1 Principle of general relativity

All physics measurements are local, since they are performed by apparati in a finite spatial and temporal extension. Therefore, by means of differential geometry, we need to write equations that take the same form in any arbitrary (local) reference frame and they must be independent of that.

Principle 1.1 (of general relativity)
The laws of physics are the same in all reference frame.

This implies that physical laws must be written in terms of tensors. Observers are physical apparati that move along trajectories in spacetime and we can cover a portion of spacetime with a reference frame starting from that.

1.2 Principles of equivalence

All fundamental forces are described by charges of opposite sign, therefore they can be neglected using zero charge objects. However, we cannot do that with gravity since it is always attractive. The only way to eliminate gravity is considering a free-falling observer.

Principle 1.2 (of equivalence (weak))
The gravitational mass m_g is equal to the inertial mass m_i.

Principle 1.3 (of equivalence (Einstein))
Motion in a uniform gravitational field is equal to free fall.

In other words, we cannot distinguish by a local experiment, if we are in a uniform gravitation field or if we are free-falling.

Principle 1.4 (of equivalence (strong))
There always exists a local reference frame in which all gravitational effects vanish.

This means that inertial frames are free-falling reference frames and they can be defined only in a small neighbourhood of a point. In fact, for these frames, the metric reduces to Minkowski and we can use Killing vectors tangent to these trajectories to build a local reference frame in each point, which is a Gaussian normal frame.

1.3 Principle of general covariance

Consequently, for inertial frames, we have $\pounds_\mu \simeq \partial_\mu \simeq \nabla_\mu$ and we restore special relativity.

Principle 1.5 (of general covariance)
The laws of physics in a general reference frame are obtained by special relativity by replacing differential geometry quantities (metric, indices, covariant derivatives).

This can be made because of the local flatness theorem.

Chapter 2

Einstein field equations

In this chapter, we will study what are test particles and what are the sources of gravity. Furthermore, we will derive the Einstein field equation.ì by physics analogy.

2.1 Test particles

Consider a massive test particle of 4-velocity $u^\mu = dx^\mu/d\tau$, which is not subjected to any force, in a free-falling frame. Therefore, we can use special relativity and write down the relativistic Newton's law

$$\frac{d^2x^\alpha}{d\tau^2} = 0 \ ,$$

where $t = \gamma\tau$ is the proper time. Moreover, in a local inertial frame $\Gamma^\alpha_{\mu\nu} \sim g_{\mu\nu,\beta} \sim \eta_{\mu\nu,\beta} = 0$, hence

$$\frac{d^2x^\alpha}{d\tau^2} + \Gamma^\alpha_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0 \ .$$

This is exactly the geodesic equation

$$u^\mu \nabla_\mu u^\alpha = 0 \ .$$

Proof. In fact,

$$u^\mu \nabla_\mu u^\alpha = \underbrace{u^\mu \partial_\mu}_{d/d\tau} u^\alpha + \Gamma^\alpha_{\mu\nu} u^\mu u^\nu = \frac{d^2x^\alpha}{d\tau^2} + \Gamma^\alpha_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0 \ .$$

q.e.d.

This derivation only applies in a point, but since it is frame independent, it implies that test particles follow geodesics. In a different frame, Christoffel symbols will not be zero and we can interpret them as a gravitational potential $\Gamma^{\alpha}_{\mu\nu} \sim g_{\mu\nu,\beta}$.

We can generalise this result also for massless particles, like light. In fact, given the light-cone property $g(\underline{u}, \underline{u})$, which is conserved along a geodesic

$$\nabla_{\underline{u}} g(\underline{u}, \underline{u}) = 0 \ ,$$

we can prove that it implies that it also follows geodesic equation

$$u^{\mu} \nabla_{\mu} u^{\alpha} = 0 \ .$$

Proof. In fact

$$\begin{aligned} 0 &= \nabla_{\underline{u}} g(\underline{u}, \underline{u}) = \nabla_{u^{\mu}\partial_{\mu}} g(\underline{u}, \underline{u}) = u^{\mu} \nabla_{\mu} g(\underline{u}, \underline{u}) \\ &= u^{\mu} g(\nabla_{\mu} \underline{u}, \underline{u}) + u^{\mu} \nabla_{\mu} g(\underline{u}, \nabla_{\mu} \underline{u}) = 2u^{\mu} g(\nabla_{\mu} \underline{u}, \underline{u}) = 2(u^{\mu} \nabla_{\mu} u^{\alpha}) u_{\alpha} \ . \end{aligned}$$

q.e.d.

Therefore, light follows geodesic, even though we cannot identify the proper time as an affine parameter.

2.2 Sources

The metric has 10 independent components, since it is a symmetric 4×4 matrix. We need 10 ten partial differential partial equations at second order. The Einstein tensor could be a valid candidate

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \ ,$$

because it is constructed by contractions of the Riemann tensor. Furthermore, it satisfies the Bianchi identity

$$\nabla_{\mu} G^{\mu\nu} = \nabla_{\nu} G^{\mu\nu} = 0 \ .$$

This implies that the actually degrees of freedom is 6.

The source part of the equation must satisfy the same properties: symmetric and Bianchi identity. By the principle of equivalence, gravitational interactions depends on proper mass, and by special relativity, mass is equivalent to energy. Therefore, the source of gravity is energy. The energy-momentum tensor is a valid candidate. For a perfect fluid, which is a fluid that depends only on proper energy density ρ and proper pressure p, with co-moving 4-velocity u^{μ}, it becomes

$$T^{\mu\nu} = (p + \rho) u^{\mu} u^{\nu} + p g^{\mu\nu} \ .$$

In matrix notation

$$T^{\mu\nu} = \begin{bmatrix} \rho & 0 \\ 0 & pg^{ij} \end{bmatrix} \ , \quad T^{\mu}{}_{\nu} = \begin{bmatrix} -\rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{bmatrix}$$

Notice that pressure is orthogonal to u^μ.

Proof. In fact,

$$(u^\mu u^\nu + g^{\mu\nu})u_\mu = u^\nu \underbrace{u^\mu u_\mu}_{-1} + u^\nu = -u^\nu + u^\nu = 0 \ .$$

q.e.d.

The 4-current $j^\mu = (\rho, p^i)$ satisfies the continuity equation

$$\partial_\mu j^\mu = 0 \ , \quad \partial_t \rho + \underline{\nabla} \cdot \underline{p} = 0 \ ,$$

which physically means that energy is conserved.

The Bianchi identity of the energy-momentum tensor can be decomposed into the conservation of energy and the Euler equation for fluids.

Proof. In fact,

$$\begin{aligned} 0 = \partial_\mu T^{\mu\nu} &= \partial_\mu((p+\rho)u^\mu u^\nu + pg^{\mu\nu}) \\ &= \partial_\mu(p+\rho)u^\mu u^\nu + (p+\rho)(\partial_\mu u^\mu)u^\nu + (p+\rho)u^\mu(\partial_\mu u^\nu) + (\partial_\mu p)g^{\mu\nu} \\ &= u^\mu u^\nu \partial_\mu(p+\rho) + (p+\rho)(u^\nu \partial_\mu u^\mu + u^\mu \partial_\mu u^\nu) + \partial^\nu p \ . \end{aligned}$$

Considering $u^\mu = (1, u^i)$, for the temporal index, we have

$$\begin{aligned} \partial_\mu T^{\mu 0} &= u^\mu \underbrace{u^0}_{1} \partial_\mu(p+\rho) + (p+\rho)(\underbrace{u^0}_{1} \partial_\mu u^\mu + u^\mu \partial_\mu \underbrace{u^0}_{1}) + \partial^0 p \\ &= u^\mu \partial_\mu(p+\rho) + (p+\rho)\partial_\mu u^\mu + \partial^0 p = \partial_\mu((p+\rho)u^\mu) - \dot{p} \\ &= \partial_0(p+\rho) + \partial_i((p+\rho)u^i) - \dot{p} \ , \end{aligned}$$

which in the non-relativistic limit $|p| \ll \rho$ becomes

$$\partial_0(p+\rho) + \partial_i((p+\rho)u^i) - \dot{p} \simeq \partial_0 \rho + \partial_i(\rho u^i) = 0 \ .$$

This is the conservation of energy for a fluid. For the spatial indices, we have

$$\begin{aligned} \partial_\mu T^{\mu i} &= u^\mu u^i \partial_\mu(p+\rho) + (p+\rho)(u^i \partial_\mu u^\mu + u^\mu \partial_\mu u^i) + \partial^i p \\ &= \underbrace{u^0}_{1} u^i \partial_0(p+\rho) + u^j u^i \partial_j(p+\rho) + (p+\rho)(u^i \partial_0 \underbrace{u^0}_{1} + \underbrace{u^0}_{1} \partial_0 u^i) \\ &\quad + (p+\rho)(u^i \partial_j u^j + u^j \partial_j u^i) + \partial^i p \\ &= u^i \partial_0(p+\rho) + u^j u^i \partial_j(p+\rho) + (p+\rho)\partial_0 u^i + (p+\rho)(u^i \partial_j u^j + u^j \partial_j u^i) + \partial^i p \\ &= u^i \underbrace{(\partial_0 \rho + \partial_j(\rho u^j))}_{0} + u^i(\partial_0 p + \partial_j(p u^j)) + (p+\rho)\partial_0 u^i + (p+\rho)(u^i \partial_j u^j + u^j \partial_j u^i) + \partial^i p \\ &= u^i(\partial_0 p + \partial_j(p u^j)) + (p+\rho)\partial_0 u^i + (p+\rho)(u^i \partial_j u^j + u^j \partial_j u^i) + \partial^i p \ , \end{aligned}$$

which in the non-relativistic limit $|p| \ll \rho$ and $u^i \ll c$ becomes

$$\rho\partial_0 u^i + \rho u^j \partial_j u^i + \partial^i p = 0 \ ,$$

This is the Euler equation for a fluid. q.e.d.

Putting together, we have

$$G_{\mu\nu} \propto T_{\mu\nu} \ .$$

The proportionality constant can be found using dimensional analysis and the Newtonian limit. Finally, we find

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ .$$

2.3 Gravity and geometry

The non-linear Einstein field equations tell us that matter curves spacetime, but it follows also geodesic. This means that gravity can be thought as geometry. For example, it can be proved that for dust, i.e. $p = 0$, that the geodesic equation follows from the continuity equation.

Proof. In fact, the energy-momentum tensor is

$$T^{\mu\nu} = \rho u^\mu u^\nu$$

and the continuity equation becomes

$$0 = \nabla_\mu T^{\mu\nu} = \nabla_\mu(\rho u^\mu u^\nu) = \nabla_\mu(\rho u^\mu u^\nu) = (\nabla_\mu \rho) u^\mu u^\nu + \rho(\nabla_\mu u^\mu) u^\nu + (\rho \nabla_\mu u^\nu) u^\mu \ .$$

Now, we multiply for u_ν and we obtain

$$\begin{aligned} 0 = (\nabla_\mu T^{\mu\nu}) u_\nu &= (\nabla_\mu \rho) u^\mu \underbrace{u^\nu u_\nu}_{-1} + \rho(\nabla_\mu u^\mu) \underbrace{u^\nu u_\nu}_{-1} + \rho(\nabla_\mu u^\nu) u^\mu u_\nu \\ &= -(\nabla_\mu \rho) u^\mu - \rho(\nabla_\mu u^\mu) + \rho \underbrace{(\nabla_\mu u^\nu) u_\nu}_{0} u^\mu = -\nabla_\mu(\rho u^\mu) \ , \end{aligned}$$

where we have used the fact that

$$0 = \rho \nabla_\mu (u_\nu u^\nu) = 2\rho(\nabla_\mu u^\nu) u_\nu \ .$$

We invert the first equation

$$\rho(\nabla_\mu u^\nu) u^\mu = -(\nabla_\mu \rho) u^\mu u^\nu - \rho(\nabla_\mu u^\mu) u^\nu = -\nabla_\mu(\rho u^\mu) u^\nu = 0 \ ,$$

which implies that

$$\rho u^\mu \nabla_\mu u^\nu = 0 \ .$$

q.e.d.

Chapter 3

Gravity and action

In this chapter, we will study how to derive the Einstein field equations by an action principle and the gauge symmetries of this action by means of diffeomorphisms.

3.1 Einstein-Hilbert action

In the vacuum, Einstein field equations can be obtained by an action, called the Einstein-Hilbert action

$$S = -\int d^4x \ \sqrt{-g}R \ .$$

Proof. We make a variation of the metric $\delta g_{\mu\nu}$, hence

$$\delta(\sqrt{-g}R) = \delta\sqrt{-g}R + \sqrt{-g}\delta R \ .$$

For the first term, we use the identity

$$\delta \det g = \delta \exp \ln \det g = \exp \ln \det g \ \delta \ln \det g = \det g \ \delta \operatorname{tr} \ln g = \det g \ \delta \operatorname{tr}(g^{-1}\delta g) \ ,$$

therefore

$$\begin{aligned} \delta\sqrt{-g} &= \frac{1}{2\sqrt{-g}}\delta\sqrt{-g} = -\frac{1}{2\sqrt{-g}}(-g) \ g^{\mu\nu}\delta g_{\mu\nu} = \frac{1}{2}\sqrt{-g} \ g^{\mu\nu}\delta g_{\mu\nu} \\ &= \frac{1}{2}\sqrt{-g} \ g^{\mu\nu}\delta g_{\mu\nu} = -\frac{1}{2}\sqrt{-g} \ g_{\mu\nu}\delta g^{\mu\nu} \ . \end{aligned} \tag{3.1}$$

For the second term,

$$\delta R = \delta(g^{\mu\nu}R_{\mu\nu}) = \delta g^{\mu\nu}R_{\mu\nu} + g^{\mu\nu}\delta R_{\mu\nu}$$

Now, we vary the Christoffel symbols

$$\delta\Gamma^{\alpha}_{\mu\nu} = \frac{1}{2}\delta(g^{\alpha\beta}(-g_{\mu\nu,\alpha} + g_{\mu\alpha,\nu} + g_{\nu\alpha,\mu})) = \frac{1}{2}g^{\alpha\beta}(-\delta g_{\mu\nu,\alpha} + \delta g_{\mu\alpha,\nu} + \delta g_{\nu\alpha,\mu}) \ . \tag{3.2}$$

Hence,

$$\delta R_{\mu\nu} = \delta\Gamma^{\alpha}_{\mu\nu;\alpha} - \delta\Gamma^{\alpha}_{\mu\alpha;\nu} \ ,$$

and

$$\begin{aligned} g^{\mu\nu}\delta R_{\mu\nu} &= g^{\mu\nu}\delta\Gamma^{\alpha}_{\mu\nu;\alpha} - \delta\Gamma^{\alpha}_{\mu\alpha;\nu} = \nabla_\alpha(g^{\mu\nu}\delta\Gamma^{\alpha}_{\mu\nu}) - \nabla_\nu(g^{\mu\nu}\delta\Gamma^{\alpha}_{\mu\alpha}) \\ &= \nabla_\alpha(g^{\mu\nu}\delta\Gamma^{\alpha}_{\mu\nu}) - \nabla_\alpha(g^{\mu\alpha}\delta\Gamma^{\nu}_{\mu\nu}) = \nabla_\alpha(g^{\mu\nu}\delta\Gamma^{\alpha}_{\mu\nu} - g^{\mu\alpha}\delta\Gamma^{\nu}_{\mu\nu}) \ , \end{aligned}$$

where we have exchanged α and ν. This term can be neglected because it is a total derivative. Putting together, we obtain

$$\delta(\sqrt{-g}R) = \sqrt{-g}\Big(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\Big)\delta g^{\mu\nu} = \sqrt{-g}G_{\mu\nu}\delta g^{\mu\nu}$$

and

$$\delta S = -\int d^4x \ \sqrt{-g} \ G_{\mu\nu}\delta g^{\mu\nu} \ .$$

Since $\delta g^{\mu\nu}$ is arbitrary, the equations of motion are

$$G_{\mu\nu} = 0 \ .$$

q.e.d.

Considering matter and dimensional analysis as well, the action becomes

$$S = \int d^4x \ \sqrt{-g}\Big(-\frac{R}{16\pi G_N} + \mathcal{L}_M\Big) = S_{EH} + S_M \ ,$$

where $\mathcal{L}_M$ is the Lagrangian density of matter.

Proof. We define the energy-momentum tensor

$$\begin{aligned} T_{\mu\nu} &= \frac{2}{\sqrt{-g}}\frac{\delta S_M}{\delta g^{\mu\nu}} = \frac{2}{\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}}(\sqrt{-g}\mathcal{L}_M) \\ &= \frac{2}{\sqrt{-g}}\sqrt{-g}\frac{\delta\mathcal{L}_M}{\delta g^{\mu\nu}} + \frac{2}{\sqrt{-g}}\mathcal{L}_M\frac{\delta\sqrt{-g}}{\delta g^{\mu\nu}} = 2\frac{\delta\mathcal{L}_M}{\delta g^{\mu\nu}} - \mathcal{L}_M g_{\mu\nu} \ , \end{aligned}$$

where we have used (3.1). Moreover, the variation of the action is

$$\delta S = \int d^4x \ \sqrt{-g}\Big(-\frac{G_{\mu\nu}}{8\pi G_N} + T_{\mu\nu})\delta g^{\mu\nu} \ ,$$

and, since $\delta g^{\mu\nu}$ is arbitrary, the equations of motion are

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ .$$

q.e.d.

3.2 Diffeomorphisms

Consider two charts $x = x^\mu(P)$ and $y = y^\mu(P)$, defined as

$$x\colon \mathcal{D}_x \to \mathbb{R}^n \ , \quad y\colon \mathcal{D}_y \to \mathbb{R}^n \ ,$$

such that they do not overlap $\mathcal{D}_x \cap \mathcal{D}_y \neq \emptyset$. A diffeomorphism is a change of coordinates that can be smoothly connected with the identity

$$y^\mu(P) = x^\mu(P) + \epsilon\xi^\mu(P) \ ,$$

where ϵ is an infinitesimal parameter and ξ^μ are components of a vector field in coordinate basis $\partial/\partial x$

$$\underline{\xi} = \xi^\mu \partial/\partial x^\mu \ .$$

The coordinate basis change as

$$\frac{\partial}{\partial y^\mu} = \frac{\partial x^\nu}{\partial y^\mu}\frac{\partial}{\partial x^\nu} = \left(\delta^\nu{}_\mu + \epsilon\frac{\partial \xi^\nu}{\partial x^\mu}\right)^{-1}\frac{\partial}{\partial x^\nu} = \left(\delta^\nu{}_\mu - \epsilon\frac{\partial \xi^\nu}{\partial x^\mu}\right)\frac{\partial}{\partial x^\nu} \ , \tag{3.3}$$

whereas the dual basis as

$$d\tilde{y}^\mu = \frac{\partial y^\mu}{\partial x^\nu} d\tilde{x}^\nu = \left(\delta^\mu{}_\nu + \epsilon\frac{\partial \xi^\mu}{\partial x^\nu}\right) d\tilde{x}^\nu \ . \tag{3.4}$$

The variation of a scalar $\phi = f \circ x^{-1}$ is

$$\delta\phi(x^\mu) = -\xi^\mu \frac{\partial \phi}{\partial x^\mu} = -\pounds_{\underline{\xi}}\phi \ .$$

This means that shifting the coordinates in the direction of $\underline{\xi}$ is equivalent to dragging backwards the scalar field f.

Proof. Consider $\phi = f \circ x^{-1}$ and $\psi = f \circ y^{-1}$. Since f is a scalar, we have

$$\psi(y^\mu) = \phi(x^\mu) \ .$$

Using the expansions

$$\psi(x^\mu) = \phi(x^\mu) + \epsilon\delta\phi(x^\mu) \ ,$$

$$\psi(y^\mu) = \psi(x^\mu + \epsilon\xi^\mu) \simeq \psi(x^\mu) + \epsilon\xi^\mu\frac{\partial\psi}{\partial x^\mu} \simeq \phi(x^\mu) + \epsilon\delta\phi(x^\mu) + \epsilon\xi^\mu\frac{\partial\phi}{\partial x^\mu} \ ,$$

we have

$$\delta\phi(x^\mu) = -\xi^\mu\frac{\partial\phi}{\partial x^\mu} \ .$$

q.e.d.

The variation of a vector $\underline{v}$ is

$$\delta v^\mu = v^\nu \frac{\partial \xi^\mu}{\partial x^\nu} - \xi^\nu \frac{\partial v^\mu}{\partial x^\nu} = -(\pounds_{\underline{\xi}} \underline{v})^\mu \ .$$

Proof. Since $\underline{v}$ is a scalar, we have

$$V^\nu(y^\mu) \frac{\partial}{\partial y^\nu} = v^\nu(x^\mu) \frac{\partial}{\partial x^\nu} \ .$$

Using (3.3) and the expansions

$$V^\nu(x^\mu) = v^\nu(x^\mu) + \epsilon \delta v^\nu(x^\mu) \ ,$$

$$\begin{aligned}
V^\nu(y^\mu) \frac{\partial}{\partial y^\nu} &= V^\nu(x^\mu + \epsilon \xi^\mu) \frac{\partial}{\partial y^\nu} \simeq \left(V^\nu(x^\mu) + \epsilon \xi^\mu \frac{\partial v^\nu}{\partial x^\mu} \right) \left(\delta^\alpha{}_\nu - \epsilon \frac{\partial \xi^\alpha}{\partial x^\nu} \right) \frac{\partial}{\partial x^\alpha} \\
&= \left(v^\nu(x^\mu) + \epsilon \delta v^\nu(x^\mu) + \epsilon \xi^\mu \frac{\partial v^\nu}{\partial x^\mu} \right) \left(\delta^\alpha{}_\nu - \epsilon \frac{\partial \xi^\alpha}{\partial x^\nu} \right) \frac{\partial}{\partial x^\alpha} \\
&= v^\nu(x^\mu) \frac{\partial}{\partial x^\nu} + \epsilon \delta v^\nu(x^\mu) \frac{\partial}{\partial x^\nu} + \epsilon \xi^\mu \frac{\partial v^\nu}{\partial x^\mu} \frac{\partial}{\partial x^\nu} - \epsilon v^\nu \frac{\partial \xi^\alpha}{\partial x^\nu} \frac{\partial}{\partial x^\alpha} \\
&= v^\nu(x^\mu) \frac{\partial}{\partial x^\nu} + \epsilon \delta v^\nu(x^\mu) \frac{\partial}{\partial x^\nu} + \epsilon \xi^\mu \frac{\partial v^\nu}{\partial x^\mu} \frac{\partial}{\partial x^\nu} - \epsilon v^\alpha \frac{\partial \xi^\nu}{\partial x^\alpha} \frac{\partial}{\partial x^\nu} \ ,
\end{aligned}$$

where we have exchanged α and ν, we have

$$\delta v^\nu(x^\mu) = v^\alpha \frac{\partial \xi^\nu}{\partial x^\alpha} - \xi^\mu \frac{\partial v^\nu}{\partial x^\mu} \ .$$

q.e.d.

The variation of the metric is

$$\delta v^\mu = v^\nu \frac{\partial \xi^\mu}{\partial x^\nu} - \xi^\nu \frac{\partial v^\mu}{\partial x^\nu} = -(\pounds_{\underline{\xi}} \underline{v})^\mu \ . \tag{3.5}$$

Proof. Since g is a scalar, we have

$$G_{\mu\nu}(y^\mu) d\tilde{y}^\mu d\tilde{y}^\nu = g_{\mu\nu}(x^\mu) d\tilde{x}^\mu d\tilde{x}^\nu \ .$$

Using (3.4) and the expansions

$$G_{\mu\nu}(x^\mu) = g_{\mu\nu}(x^\mu) + \epsilon \delta g_{\mu\nu}(x^\mu) \ ,$$

$$\begin{aligned}
G_{\mu\nu}(y^\mu) d\tilde{y}^\mu d\tilde{y}^\nu &= G_{\mu\nu}(x^\mu + \epsilon \xi^\mu) d\tilde{y}^\mu d\tilde{y}^\nu \\
&= (G_{\mu\nu}(x^\mu) + \epsilon \xi^\alpha g_{\mu\nu,\alpha})(\delta^\mu{}_\alpha + \epsilon \xi^\mu{}_{,\alpha})(\delta^\nu{}_\beta + \epsilon \xi^\nu{}_{,\beta}) d\tilde{x}^\alpha d\tilde{x}^\beta \\
&= (g_{\mu\nu}(x^\mu) + \epsilon \delta g_{\mu\nu}(x^\mu) + \epsilon \xi^\alpha g_{\mu\nu,\alpha})(\delta^\mu{}_\alpha + \epsilon \xi^\mu{}_{,\alpha})(\delta^\nu{}_\beta + \epsilon \xi^\nu{}_{,\beta}) d\tilde{x}^\alpha d\tilde{x}^\beta \\
&= g_{\alpha\beta} d\tilde{x}^\alpha d\tilde{x}^\beta + \epsilon g_{\alpha\nu} \xi^\nu{}_{,\beta} d\tilde{x}^\alpha d\tilde{x}^\beta + \epsilon g_{\mu\beta} \xi^\mu{}_{,\alpha} d\tilde{x}^\alpha d\tilde{x}^\beta \\
&\quad + \epsilon \delta g_{\alpha\beta}(x^\mu) d\tilde{x}^\alpha d\tilde{x}^\beta + \epsilon \xi^\sigma g_{\alpha\beta,\sigma} d\tilde{x}^\alpha d\tilde{x}^\beta \ ,
\end{aligned}$$

we have

$$\begin{aligned}
\delta g_{\alpha\beta}(x^\mu) &= -g_{\alpha\nu}\xi^\nu{}_{,\beta} - g_{\mu\beta}\xi^\mu{}_{,\alpha} - \xi^\sigma g_{\alpha\beta,\sigma} \\
&= -\partial_\beta(g_{\alpha\nu}\xi^\nu) - \partial_\alpha(g_{\mu\beta}\xi^\mu) + g_{\alpha\nu,\beta}\xi^\nu + g_{\mu\beta,\alpha}\xi^\mu - \xi^\sigma g_{\alpha\beta,\sigma} \\
&= -\xi_{\alpha,\beta} - \xi_{\beta,\alpha} + g_{\alpha\nu,\beta}\xi^\nu + g_{\mu\beta,\alpha}\xi^\mu - \xi^\sigma g_{\alpha\beta,\sigma} \\
&= -\xi_{\alpha,\beta} - \xi_{\beta,\alpha} + \xi^\sigma(-g_{\alpha\beta,\sigma} + g_{\alpha\sigma,\beta} + g_{\beta\sigma,\alpha}) \\
&= -\xi_{\alpha,\beta} - \xi_{\beta,\alpha} + \xi_\rho \underbrace{g^{\rho\sigma}(-g_{\alpha\beta,\sigma} + g_{\alpha\sigma,\beta} + g_{\beta\sigma,\alpha})}_{2\Gamma^\rho_{\alpha\beta}} \\
&= -\xi_{\alpha,\beta} - \xi_{\beta,\alpha} + 2\xi_\rho\Gamma^\rho_{\alpha\beta} = -\xi_{\alpha,\beta} + \xi_\rho\Gamma^\rho_{\alpha\beta} - \xi_{\beta,\alpha} + \xi_\rho\Gamma^\rho_{\beta\alpha} = -\xi_{\alpha;\beta} - \xi_{\beta;\alpha} \ ,
\end{aligned}$$

where we have integrated by parts and used the symmetric properties. q.e.d.

3.3 Gauge symmetries

The Einstein-Hilbert action is invariant under these diffeomorphisms. The vacuum action vanishes identically due to the Bianchi identity

$$S_{EH} = \int d^4x \ \sqrt{-g}R \ .$$

Proof. In fact, using (3.5)

$$\begin{aligned}
\delta S_{EH} &= \int d^4x \ \sqrt{-g}G^{\mu\nu}\delta g_{\mu\nu} = -\int d^4x \ \sqrt{-g}G^{\mu\nu}(\nabla_\nu\xi_\mu + \nabla_\mu\xi_\nu) \\
&= \int d^4x \ \sqrt{-g}(\xi_\mu \underbrace{\nabla_\nu G^{\mu\nu}}_{0} + \xi_\nu \underbrace{\nabla_\mu G^{\mu\nu}}_{0}) = 0 \ ,
\end{aligned}$$

where we have integrated by parts. q.e.d.

The matter action vanishes identically due to the continuity equation

$$S_{EH} = \int d^4x \ \sqrt{-g}\mathcal{L}_M \ .$$

Proof. In fact, using (3.5)

$$\begin{aligned}
\delta S_{EH} &= \int d^4x \ \sqrt{-g}T^{\mu\nu}\delta g_{\mu\nu} = -\int d^4x \ \sqrt{-g}T^{\mu\nu}(\nabla_\nu\xi_\mu + \nabla_\mu\xi_\nu) \\
&= \int d^4x \ \sqrt{-g}(\xi_\mu \underbrace{\nabla_\nu T^{\mu\nu}}_{0} + \xi_\nu \underbrace{\nabla_\mu T^{\mu\nu}}_{0}) = 0 \ ,
\end{aligned}$$

where we have integrated by parts. q.e.d.

Furthermore, the Riemann tensor is invariant under diffeomorphisms.

Proof. In fact, using (3.2) and (3.5),

$$
\begin{aligned}
\delta R^{\mu}{}_{\nu\alpha\beta} &= -\delta\Gamma^{\mu}_{\nu\alpha,\beta} + \delta\Gamma^{\mu}_{\nu\beta,\alpha} \\
&= -\frac{1}{2}g^{\mu\rho}\Big(- \delta g_{\nu\alpha,\rho\beta} + \delta g_{\nu\rho,\alpha\beta} + \delta g_{\alpha\rho,\nu\beta}\Big) + \frac{1}{2}g^{\mu\rho}\Big(- \delta g_{\nu\beta,\rho\alpha} + \delta g_{\nu\rho,\beta\alpha} + \delta g_{\beta\rho,\nu\alpha}\Big) \\
&= \frac{1}{2}g^{\mu\rho}\Big(\delta g_{\nu\alpha,\rho\beta} - \delta g_{\nu\rho,\alpha\beta} - \delta g_{\alpha\rho,\nu\beta} - \delta g_{\nu\beta,\rho\alpha} + \delta g_{\nu\rho,\beta\alpha} + \delta g_{\beta\rho,\nu\alpha}\Big) \\
&= -\frac{1}{2}g^{\mu\rho}\Big(\cancel{\xi_{\nu,\alpha\rho\beta}} + \cancel{\xi_{\alpha,\nu\rho\beta}} - \cancel{\xi_{\nu,\rho\alpha\beta}} - \cancel{\xi_{\rho,\nu\alpha\beta}} - \cancel{\xi_{\alpha,\rho\nu\beta}} - \cancel{\xi_{\rho,\alpha\beta\nu}} \\
&\qquad - \cancel{\xi_{\nu,\beta\rho\alpha}} - \cancel{\xi_{\beta,\nu\rho\alpha}} + \cancel{\xi_{\nu,\rho\beta\alpha}} + \cancel{\xi_{\rho,\nu\beta\alpha}} + \cancel{\xi_{\beta,\rho\nu\alpha}} + \cancel{\xi_{\rho,\beta\nu\alpha}}\Big) = 0 \ ,
\end{aligned}
$$

where we have used the fact that partial derivatives commute. q.e.d.

Part II

Gravitational waves

Chapter 4

Linearised Einstein field equations

In this chapter, we will study the weak-field limit, in order to find the linearised Einstein field equations in the De Donder gauge. With the addition of non-relativistic limit, we will recover the experimentally tested Newtonian laws (gravity and second law of dynamics).

4.1 Linearised Einstein field equations

Consider the weak field limit, in which local curvature is small. This means that there exists a reference frame such that

$$g_{\mu\nu} = \eta_{\mu\nu} + \epsilon h_{\mu\nu} \ ,$$

where ϵ is an infinitesimal parameter. The linearised Einstein field equations in first order in ϵ are

$$\eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta} = 16\pi G_N \epsilon T_{\mu\nu} \ .$$

Proof. The Christoffel symbols in first order in ϵ are

$$\begin{aligned}
\Gamma^\alpha_{\mu\nu} &= \frac{1}{2}g^{\alpha\beta}(g_{\mu\beta,\nu} + g_{\nu\beta,\mu} - g_{\mu\nu,\beta}) \\
&= \frac{1}{2}g^{\alpha\beta}(\cancel{\eta_{\mu\beta,\nu}} + \epsilon h_{\mu\beta,\nu} + \cancel{\eta_{\nu\beta,\mu}} + \epsilon h_{\nu\beta,\mu} - \cancel{\eta_{\mu\nu,\beta}} - \epsilon h_{\mu\nu,\beta}) \\
&= \frac{\epsilon}{2}g^{\alpha\beta}(h_{\mu\beta,\nu} + h_{\nu\beta,\mu} - h_{\mu\nu,\beta}) \\
&\simeq \frac{\epsilon}{2}(\eta^{\alpha\beta} - \epsilon h^{\alpha\beta})(h_{\mu\beta,\nu} + h_{\nu\beta,\mu} - h_{\mu\nu,\beta}) \\
&\simeq \frac{1}{2}(\epsilon\eta^{\alpha\beta} - \cancel{\epsilon^2 h^{\alpha\beta}})(h_{\mu\beta,\nu} + h_{\nu\beta,\mu} - h_{\mu\nu,\beta}) \\
&\simeq \frac{\epsilon}{2}\eta^{\alpha\beta}(h_{\mu\beta,\nu} + h_{\nu\beta,\mu} - h_{\mu\nu,\beta}) \ ,
\end{aligned} \tag{4.1}$$

where we have Taylor expanded the inverse of the metric

$$g^{\alpha\beta} = (g_{\alpha\beta})^{-1} = (\eta_{\alpha\beta} + \epsilon h_{\alpha\beta})^{-1} \simeq \eta_{\alpha\beta} - \epsilon h_{\alpha\beta} \ .$$

The Riemann tensor in first order in ϵ is

$$\begin{aligned}
R^{\mu}{}_{\nu\alpha\beta} &= \Gamma^{\mu}_{\nu\beta,\alpha} - \Gamma^{\mu}_{\nu\alpha,\beta} + \Gamma^{\lambda}_{\nu\beta}\Gamma^{\mu}_{\lambda\alpha} - \Gamma^{\lambda}_{\nu\alpha}\Gamma^{\mu}_{\lambda\beta} \\
&\simeq \Gamma^{\mu}_{\nu\beta,\alpha} - \Gamma^{\mu}_{\nu\alpha,\beta} \\
&= \partial_\alpha(\frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\nu\lambda,\beta} + h_{\beta\lambda,\nu} - h_{\nu\beta,\lambda})) - \partial_\beta(\frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\nu\lambda,\alpha} + h_{\alpha\lambda,\nu} - h_{\nu\alpha,\lambda})) \\
&= \frac{\epsilon}{2}\cancel{\eta^{\mu\lambda}{}_{,\alpha}}(h_{\nu\lambda,\beta} + h_{\beta\lambda,\nu} - h_{\nu\beta,\lambda}) + \frac{\epsilon}{2}\eta^{\mu\lambda}(\cancel{h_{\nu\lambda,\beta\alpha}} + h_{\beta\lambda,\nu\alpha} - h_{\nu\beta,\lambda\alpha}) \\
&\quad - \frac{\epsilon}{2}\cancel{\eta^{\mu\lambda}{}_{,\beta}}(h_{\nu\lambda,\alpha} + h_{\alpha\lambda,\nu} - h_{\nu\alpha,\lambda}) - \frac{\epsilon}{2}\eta^{\mu\lambda}(\cancel{h_{\nu\lambda,\alpha\beta}} + h_{\alpha\lambda,\nu\beta} - h_{\nu\alpha,\lambda\beta}) \\
&= \frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\beta\lambda,\nu\alpha} - h_{\nu\beta,\lambda\alpha}) - \frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\alpha\lambda,\nu\beta} - h_{\nu\alpha,\lambda\beta}) \\
&= \frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\beta\lambda,\nu\alpha} - h_{\nu\beta,\lambda\alpha} - h_{\alpha\lambda,\nu\beta} + h_{\nu\alpha,\lambda\beta}) \ .
\end{aligned}$$

The Ricci tensor in first order in ϵ is

$$\begin{aligned}
R_{\nu\beta} &= R^{\mu}{}_{\nu\mu\beta} \\
&= \frac{\epsilon}{2}\eta^{\mu\lambda}(h_{\beta\lambda,\nu\mu} - h_{\nu\beta,\lambda\mu} - h_{\mu\lambda,\nu\beta} + h_{\nu\mu,\lambda\beta}) \\
&= \frac{\epsilon}{2}\eta^{\mu\lambda}(\partial_\nu\partial_\mu h_{\beta\lambda} - \partial_\lambda\partial_\mu h_{\nu\beta} - \partial_\nu\partial_\beta h_{\mu\lambda} + \partial_\lambda\partial_\beta h_{\nu\mu}) \\
&= \frac{\epsilon}{2}(\partial_\nu\partial^\mu h_{\beta\mu} - \partial^\mu\partial_\mu h_{\nu\beta} - \partial_\nu\partial_\beta h^{\mu}{}_{\mu} + \partial^\mu\partial_\beta h_{\nu\mu}) \\
&= \frac{\epsilon}{2}(\partial^\mu\partial_\beta h_{\nu\mu} + \partial^\mu\partial_\nu h_{\beta\mu} - \partial_\nu\partial_\beta h - \Box h_{\nu\beta}) \ ,
\end{aligned}$$

where we have set $\lambda = \mu$. The Ricci scalar in first order in ϵ is

$$\begin{aligned}
R &= R^{\nu}{}_{\nu} \\
&= \eta^{\nu\beta}R_{\nu\beta} \\
&= \frac{\epsilon}{2}\eta^{\nu\beta}(\partial^\mu\partial_\beta h_{\nu\mu} + \partial^\mu\partial_\nu h_{\beta\mu} - \partial_\nu\partial_\beta h - \Box h_{\nu\beta}) \\
&= \frac{\epsilon}{2}(\partial^\mu\partial^\nu h_{\nu\mu} + \partial^\mu\partial^\beta h_{\beta\mu} - \partial_\nu\partial^\nu h - \Box h^{\nu}{}_{\nu}) \\
&= \frac{\epsilon}{2}(2\partial^\mu\partial^\nu h_{\nu\mu} - 2\Box h) \\
&= \epsilon(\partial^\mu\partial^\nu h_{\nu\mu} - \Box h) \ ,
\end{aligned}$$

where we have set $\beta = \nu$. Finally, the Einstein tensor in first order in ϵ is

$$\begin{aligned}
G_{\mu\nu} &= R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} \\
&= \frac{\epsilon}{2}(\partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta} - \partial_\mu\partial_\nu h - \Box h_{\mu\nu}) - \frac{1}{2}\epsilon(\partial^\alpha\partial^\beta h_{\alpha\beta} - \Box h)g_{\mu\nu} \\
&= \frac{\epsilon}{2}(\eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta}) \ .
\end{aligned}$$

The linearised Einstein field equations are

$$G_{\mu\nu} = 8\pi G_N T_{\mu\nu} \ ,$$

$$\frac{\epsilon}{2}(\eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta}) = 8\pi G_N \epsilon\epsilon T_{\mu\nu} \ ,$$

$$\eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta} = 16\pi G_N \epsilon T_{\mu\nu} \ ,$$

where we have expanded ${T'}_{\mu\nu} = T^{(0)}_{\mu\nu} + \epsilon T_{\mu\nu}$ and noticed that $T^{(0)}_{\mu\nu} = 0$, because the Minkowski metric solves Einstein equations. q.e.d.

Since the metric is invariant under a gauge symmetry, following by the Bianchi identity, we can choose a suitable gauge fixing to simplify these equations: the De Donder gauge

$$\partial^\mu h_{\mu\nu} = \frac{1}{2}\partial_\nu h \ .$$

Therefore, the linearised Einstein field equations becomes

$$-\Box h_{\mu\nu} = 16\pi G_N (T_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} T) \ .$$

Proof. In fact, the right side

$$\begin{aligned}
G_{\mu\nu} &= \eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta} \\
&= \eta_{\mu\nu}\partial^\alpha\partial_\alpha h - \partial^\alpha\partial_\alpha h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta} \\
&= \eta_{\mu\nu}\partial^\alpha\partial_\alpha h - \partial^\alpha\partial_\alpha h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha \underbrace{\partial^\beta h_{\beta\alpha}}_{\frac{1}{2}\partial_\alpha h} + \partial_\nu \underbrace{\partial^\beta h_{\beta\mu}}_{\frac{1}{2}\partial_\mu h} + \partial_\mu \underbrace{\partial^\beta h_{\beta\nu}}_{\frac{1}{2}\partial_\nu h} \\
&= \eta_{\mu\nu}\partial^\alpha\partial_\alpha h - \partial^\alpha\partial_\alpha h_{\mu\nu} - \partial_\mu\partial_\nu h - \frac{1}{2}\eta_{\mu\nu}\partial^\alpha\partial_\alpha h + \frac{1}{2}\partial_\mu h + \frac{1}{2}\partial_\mu\partial_\nu h \\
&= \eta_{\mu\nu}\partial^\alpha\partial_\alpha h - \partial^\alpha\partial_\alpha h_{\mu\nu} - \cancel{\partial_\mu\partial_\nu h} - \frac{1}{2}\eta_{\mu\nu}\partial^\alpha\partial_\alpha h + \cancel{\frac{1}{2}\partial_\nu\partial_\mu h} + \cancel{\frac{1}{2}\partial_\mu\partial_\nu h} \\
&= -\Box h_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}\Box h \ .
\end{aligned}$$

Hence,

$$\Box h_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}\Box h = 16\pi G_N T_{\mu\nu} \ .$$

Furthermore, the trace of the Einstein tensor is

$$\begin{aligned}
\eta^{\mu\nu}G_{\mu\nu} &= \eta^{\mu\nu}(\eta_{\mu\nu}\Box h - \Box h_{\mu\nu} - \partial_\mu\partial_\nu h - \eta_{\mu\nu}\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial_\nu h_{\mu\beta} + \partial^\beta\partial_\mu h_{\nu\beta}) \\
&= \underbrace{\eta^{\mu\nu}\eta_{\mu\nu}}_{4}\Box h - \Box\underbrace{\eta^{\mu\nu}h_{\mu\nu}}_{h} - \eta^{\mu\nu}\partial_\mu\partial_\nu h - \underbrace{\eta^{\mu\nu}\eta_{\mu\nu}}_{4}\partial^\alpha\partial^\beta h_{\alpha\beta} + \eta^{\mu\nu}\partial^\beta\partial_\nu h_{\mu\beta} + \eta^{\mu\nu}\partial^\beta\partial_\mu h_{\nu\beta} \\
&= 4\Box h - \Box h - \partial^\nu\partial_\nu h - 4\partial^\alpha\partial^\beta h_{\alpha\beta} + \partial^\beta\partial^\mu h_{\mu\beta} + \partial^\beta\partial^\nu h_{\nu\beta} \\
&= 2\Box h - 2\partial^\nu\underbrace{\partial^\mu h_{\mu\nu}}_{\frac{1}{2}\partial_\nu h} \\
&= \Box h \ ,
\end{aligned}$$

which is equal to

$$16\pi G_N \underbrace{\eta^{\mu\nu}T_{\mu\nu}}_{T} = 16\pi G_N T \ .$$

Putting together

$$\begin{aligned}
-\Box h_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}\Box h &= 16\pi G_N T_{\mu\nu} \ , \\
-\Box h_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}(16\pi G_N T) &= 16\pi G_N T_{\mu\nu} \ , \\
-\Box h_{\mu\nu} &= 16\pi G_N\Big(T_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}T) \ .
\end{aligned}$$

q.e.d.

4.2 Newtonian regime

In order to recover the full Newtonian regime, we need also to ask that the matter source is static and the only non-vanishing component of the energy momentum tensor is

$$T^{\mu\nu} \simeq T_{00} = -T \ .$$

Therefore, we recover the Poisson equation

$$\nabla^2 V_N = 4\pi G_N \rho$$

where $h_{00} = -2V_N$ and G_N is indeed the Newton constant.

Proof. In fact

$$T^{00}(\underline{x}) = (\rho + p)u^0u^0 + p\eta^{00} = \rho(\underline{x}) \ .$$

Hence,

$$\Box h_{00}(\underline{x}) \simeq \nabla^2 h_{00}(\underline{x}) = -8\pi G_N T_{00}(\underline{x}) = -8\pi G_N \rho(\underline{x}) \ .$$

We identify $h_{00} = -2V_N$ and we obtain

$$\nabla^2 V_N = 4\pi G_N \rho \ .$$

q.e.d.

Furthermore, if we consider the non-relativistic limit, in which particles are not static, but move slower $u^i = \epsilon v^i \ll c$. The geodesic equation becomes

$$\frac{d^2x^i}{dt^2} \simeq -\frac{\partial V_N}{\partial x^i} \ .$$

Proof. We expand the 4-velocity $u^\mu = (1, \epsilon v^i)$ so that

$$\frac{d^2x^\mu}{d\tau^2} = \epsilon\Big(0, \frac{dv^i}{dt}\Big) = \epsilon\Big(0, \frac{d^2x^i}{dt^2}\Big) \ .$$

Using (4.1), we find

$$\begin{aligned}
\Gamma^\beta_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} &= \frac{\epsilon}{2}\eta^{\alpha\beta}(-h_{\mu\nu,\alpha} + h_{\mu\alpha,\nu} + h_{\nu\alpha,\mu})\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} \\
&= \frac{\epsilon}{2}\eta^{\alpha\beta}(-h_{\mu\nu,\alpha} + h_{\mu\alpha,\nu} + h_{\nu\alpha,\mu})\delta^\mu{}_0\delta^\nu{}_0 \\
&= \frac{\epsilon}{2}\eta^{\alpha\beta}(-h_{00,\alpha} + \underbrace{h_{0\alpha,0}}_{0} + \underbrace{h_{0\alpha,0}}_{0}) \\
&= -\frac{\epsilon}{2}\eta^{\alpha\beta}h_{00,\alpha}
\end{aligned}$$

where we have neglected terms of higher order. Putting together

$$\frac{d^2x^\alpha}{d\tau^2} + \Gamma^\alpha\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} \simeq \epsilon\Big(\frac{d^2x^i}{dt^2} + \frac{1}{2}h_{00,i}\Big)$$

and using $h_{00} = -2V_N$, we find

$$\frac{d^2x^i}{dt^2} \simeq -\frac{\partial V_N}{\partial x^i} \ .$$

q.e.d.

Chapter 5

Gravitational waves

In this chapter, we will study how to solve the linearised Einstein field equations with gravitational plane waves: form, production and detection of them.

5.1 Plane waves in vacuum

In is useful to express the perturbation part of the metric tensor h, in terms of the transverse tensor

$$h^T_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h \ .$$

It is equivalent, because once we find the transverse tensor, we can always recover the original one. The De Donder gauge becomes the transverse gauge

$$\partial^\mu h^T_{\mu\nu} = 0 \ .$$

It is also called the trace-inverse, since

$$\eta^{\mu\nu}h^T_{\mu\nu} = \eta^{\mu\nu}h_{\mu\nu} - \frac{1}{2}\underbrace{\eta^{\mu\nu}\eta_{\mu\nu}}_{4}h = h - 2h = -h \ .$$

The linearised Einstein field equations becomes

$$-\Box h^T_{\mu\nu} = 16\pi G_N T_{\mu\nu} \ .$$

Proof. In fact

$$16\pi G_N T_{\mu\nu} = -\Box h_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}\Box h = -\Box\Big(h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h\Big) = -\Box h^T_{\mu\nu} \ .$$

q.e.d.

Now, we consider the vacuum case. For massless gravitational waves, the equations of motion become

$$\Box h^T_{\mu\nu} = 0 \ .$$

However, we notice that there is still a residual gauge, always inside the De Donder one. In fact, if we make another diffeomorphism transformation

$$\tilde{h}_{\mu\nu} = h_{\mu\nu} + \xi_{\mu,\nu} + \xi_{\nu,\mu} \ ,$$

where ξ^μ is a harmonic, i.e. $\Box\xi^\mu = 0$, we find that both the equations of motion and the transverse gauge are still satisfied.

Proof. For the equations of motion, we have

$$\begin{aligned}
\Box\tilde{h}^T_{\mu\nu} &= \Box\tilde{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\Box\tilde{h} \\
&= \Box h_{\mu\nu} + \underbrace{\Box\xi_{\mu,\nu}}_{0} + \underbrace{\Box\xi_{\nu,\mu}}_{0} - \frac{1}{2}\eta_{\mu\nu}\Box h - \frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\underbrace{\Box\xi_{\alpha,\beta}}_{0} - \frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\underbrace{\Box\xi_{\beta,\alpha}}_{0} \\
&= \Box h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\Box h = \Box h^T_{\mu\nu} = 0 \ .
\end{aligned}$$

For the transverse gauge, we have

$$\begin{aligned}
\partial^\mu\tilde{h}^T_{\mu\nu} &= \partial^\mu\tilde{h}_{\mu\nu} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\tilde{h} \\
&= \partial^\mu h_{\mu\nu} + \partial^\mu\xi_{\mu,\nu} + \partial^\mu\xi_{\nu,\mu} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}h - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\alpha,\beta} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\beta,\alpha} \\
&= \underbrace{\partial^\mu h_{\mu\nu} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}h}_{0} + \partial^\mu\xi_{\mu,\nu} + \partial^\mu\xi_{\nu,\mu} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\alpha,\beta} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\beta,\alpha} \\
&= \partial^\mu\xi_{\mu,\nu} + \partial^\mu\xi_{\nu,\mu} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\alpha,\beta} - \partial^\mu\frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}\xi_{\beta,\alpha} \\
&= \eta^{\alpha\beta}\left(\cancel{\xi_{\beta,\nu\alpha}} + \xi_{\nu,\beta\alpha} - \cancel{\frac{1}{2}\xi_{\alpha,\beta\nu}} - \cancel{\frac{1}{2}\xi_{\beta,\alpha\nu}}\right) = \Box\xi_\nu = 0 \ .
\end{aligned}$$

q.e.d.

Now, consider a plane wave ansatz

$$h^T_{\mu\nu}(x) = \epsilon_{\mu\nu}\exp(ik_\alpha x^\alpha) \ ,$$

where $\epsilon_{\mu\nu}$ are the components of a symmetric polarisation tensor

$$\epsilon_{\mu\nu} =$$

Using the equations of motion, we find the dispersion relation for a massless wave

$$k^\alpha k_\alpha = 0 \ ,$$

which shows that gravitational waves travel at the speed of light.

Proof. In fact,

$$\Box h^T_{\mu\nu} = -k^\alpha k_\alpha h^T_{\mu\nu} = 0 \ .$$

q.e.d.

Using the transverse gauge, we find the transversality condition

$$k^\mu \epsilon_{\mu\nu} = k^\nu \epsilon_{\mu\nu} = 0 \ .$$

Proof. In fact,

$$\partial^\mu h^T_{\mu\nu} = ik^\mu \epsilon_\mu \exp(ik^\alpha x_\alpha) = 0 \ .$$

q.e.d.

Consider a plane wave moving along the z-axis, with wave vector $k^\mu = (k, 0, 0, k)$. This implies that for a generic vector v

$$0 = k_\mu v^\mu = k(v^0 - v^3) \ , \quad v^0 = v^3 \ .$$

Therefore, the components of the polarisation tensor are now

$$\epsilon_{\mu\nu} = \begin{bmatrix} a & b & c & a \\ b & e & f & b \\ c & f & h & c \\ a & b & c & a \end{bmatrix} \ ,$$

where we have used $\epsilon_{0\mu} = \epsilon_{\mu 0} = \epsilon_{3\mu} = \epsilon_{\mu 3}$. However, we still have a residual gauge freedom to satisfies: we choose 3 degrees of freedom to put $a = b = c = 0$ and the remaining one to set the trace equals to zero $e + f = 0$. Hence

$$\epsilon_{\mu\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & e & f & 0 \\ 0 & f & -e & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \ .$$

Putting together, we have

$$h^{TT}_{\mu\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & e & f & 0 \\ 0 & f & -e & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \exp(-ikt + ikz) \ .$$

Now, we can write the general gravitational plane wave as linear combination of

$$\begin{aligned}
h^{TT}_{\mu\nu} &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & e & f & 0 \\ 0 & f & -e & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \exp(-ikt + ikz) \\
&= e \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \exp(-ikt + ikz) + f \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \exp(-ikt + ikz) \\
&= e\epsilon_+ \exp(-ikt + ikz) + f\epsilon_\times \exp(-ikt + ikz) \ ,
\end{aligned}$$

where we have defined the

$$\epsilon_+ = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \ , \quad \epsilon_\times = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \ .$$

Now, we take the real part, and we find

$$\begin{aligned}
\mathrm{Re}\, h^{TT}_{\mu\nu} &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & e\sin(kt - kz) & 0 & 0 \\ 0 & 0 & -e\sin(kt - kz) & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & f\sin(kt - kz) & 0 \\ 0 & f\sin(kt - kz) & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\
&= \mathrm{Re}\, h^{TT}_{\mu\nu\ +} + \mathrm{Re}\, h^{TT}_{\mu\nu\ \times} \ .
\end{aligned}$$

Consider now a free-falling test particle located at $x = 0$ in space, we find that a circle will turn into an ellipse along x and y

$$ds^2_+ = (\eta_{\mu\nu} + \mathrm{Re}\, h^{TT}_{\mu\nu\ +})dx^\mu dx^\nu = -dt^2 + (1 + e\sin(kt))dx^2 + (1 - e\sin(kt))dy^2 + dz^2 \ ,$$

or diagonally

$$ds^2_\times = (\eta_{\mu\nu} + \mathrm{Re}\, h^{TT}_{\mu\nu\ \times})dx^\mu dx^\nu = -dt^2 + dx^2 + dy^2 + 2f\sin(kt)dxdy + dz^2 \ .$$

We can show that trajectories at constant x, y and $z = 0$ are indeed geodesics.

Proof. In fact, consider the Lagrangian for a particle of mass m of the metric g_+

$$L = \frac{m}{2}\Big(- \dot{t}^2 + (1 + a\sin(kt))\dot{x}^2 + (1 - a\sin(kt))\dot{x}^2\Big)\dot{y}^2 + \dot{z}^2 \ .$$

Since L does not depend on x, y and z, their momenta are conserved and they can be set to zero with suitable initial conditions. Furthermore, because $L = -m/2$, we have $\dot{t}^2 = 1$, which solved means $t = \tau + t_0$. We have built a locally inertial reference frame (t, x, y, z). q.e.d.

However, their velocity does not vanish.

Proof. In fact, for a particle located in $x = l_0$ and $y = z = 0$, we have

$$l = \int_0^{l_0} dx g_{xx} = \int_0^{l_0} dx \sqrt{1 + e\sin(kt)} = \sqrt{1 + e\sin(kt)} l_0 \simeq (1 + \frac{e}{2}\sin(kt)) l_0 \ ,$$

where we have used $e \ll 1$ for weak field. Therefore, the velocity is

$$v = \frac{dl}{dt} \simeq \frac{e}{2} k \cos(kt) l_0 \ .$$

q.e.d.

5.2 Production

Consider four particles of mass m on the plane xy force to oscillate as a gravitational wave. In locally inertial frame, we have

$$x_1 = -x_2 = l_0 + \Delta l \sin(\omega t) \ ,$$

$$y_3 = -y_4 = l_0 - \Delta l \sin(\omega t) \ ,$$

$$y_1 = y_2 = x_3 = x_4 = 0 \ ,$$

where $i = 1, 2, 3, 4$. The center of mass is fixed and the total mass of the system is $M = 4m$. The total dipole moment is

$$D = m\Big(\sum_{i=1}^{2} x_i(t) + \sum_{i=3}^{4} y_i(t)\Big) = 0 \ ,$$

whereas the quadrupole moment is

$$Q(t) = m\Big(\sum_{i=1}^{2} x_i^2(t) + \sum_{i=3}^{4} y_i^2(t)\Big) = M(l_0 + \Delta l \sin(\omega t))^2 \ .$$

The total kinetic energy is

$$E = \frac{m}{2}\Big(\sum_{i=1}^{2} \dot{x}_i^2(t) + \sum_{i=3}^{4} \dot{y}_i^2(t)\Big) = \frac{M}{2}\Delta l^2 \omega^2 \cos^2(\omega t) \ .$$

Comparing it with the energy of a gravitational wave, we find

$$E_{gw} = \frac{1}{2}(4m)v^2 = \frac{M}{8} l_0^2 a^2 k^2 \sin^2(kt + t_0) \ ,$$

hence $\omega = k$, $t_0 = \pi/2$ and the strain is $a \sim \delta l / l_0 \sim \Delta l / l$.

5.3 Quadrupole formula

We can make this analysis more formally. Consider the linearised Einstein equation in transverse gauge

$$-\Box h^T_{\mu\nu} = 16\pi G_N T_{\mu\nu} \ .$$

The solution for this equation can be obtained by means of the Green function $G(x-y)$, which satisfies the Green equation

$$\Box_x G(x-y) = \delta^4(x-y) \ ,$$

where x and y are spacetime 4-vectors. The general solution of the Einstein equations is given by

$$h^T_{\mu\nu} = -16\pi G_N \int d^4y \ G(x-y) T_{\mu\nu}(y) \ .$$

where we have ignored the factor $\sqrt{-g}$ since we are linear. Green function depends on the boundary conditions. We choose the retarded Green function, since we want to study the effects of signal after it is propagated in the past. The Green function of the d'Alambertian is given by

$$G(x-y) = -\frac{1}{4\pi|\underline{x}-\underline{y}|}\theta(x^0-y^0)\delta(|\underline{x}-\underline{y}|-(x^0-y^0)) \ .$$

Therefore,

$$\begin{aligned} h^T_{\mu\nu} &= -16\pi G_N \int d^4y \ G(x-y) T_{\mu\nu}(y) \\ &= -16\pi G_N \int d^4y \ T_{\mu\nu}(y)\Big(-\frac{1}{4\pi|\underline{x}-\underline{y}|}\theta(x^0-y^0)\delta(|\underline{x}-\underline{y}|-(x^0-y^0))\Big) \\ &= 4G_N \int d^3y \ \frac{1}{|\underline{x}-\underline{y}|} T_{\mu\nu}(t_r, \underline{y}) \ , \end{aligned}$$

where $t_r = t - |\underline{x}-\underline{y}|$ is the retarded time. Physically, it means that the effect of the gravitational disturbance at a point $t, \underline{x}$ can be traced back to points on the past light cone $(t_r, \underline{x}-\underline{y})$. Now, we take the Fourier transform of the metric perturbation

$$\begin{aligned} \tilde{h}^T_{\mu\nu}(\omega, \underline{x}) &= \int \frac{dt}{(2\pi)^{1/2}} \exp(i\omega t) h^T_{\mu\nu}(t, \underline{x}) \\ &= 4G_N \int \frac{dt}{(2\pi)^{1/2}} d^3y \ \exp(i\omega t) \frac{1}{|\underline{x}-\underline{y}|} T_{\mu\nu}(t_r, \underline{y}) \end{aligned}$$

We make a change of variable into t_r and we find

$$\begin{aligned}\tilde{h}^T_{\mu\nu}(\omega,\underline{x}) &= 4G_N \int \frac{dt_r}{(2\pi)^{1/2}} d^3y \ \exp(i\omega(t_r+|\underline{x}-\underline{y}|))\frac{1}{|\underline{x}-\underline{y}|}T_{\mu\nu}(t_r,\underline{y}) \\ &= 4G_N \underbrace{\int \frac{dt_r}{(2\pi)^{1/2}} \ \exp(i\omega t_r)T_{\mu\nu}(t_r,\underline{y})}_{\tilde{T}_{\mu\nu}(\omega,\underline{y})} \int d^3y \exp(i\omega|\underline{x}-\underline{y}|)\frac{1}{|\underline{x}-\underline{y}|} \\ &= 4G_N \int d^3y \ \exp(i\omega|\underline{x}-\underline{y}|)\frac{1}{|\underline{x}-\underline{y}|}\tilde{T}_{\mu\nu}(\omega,\underline{y}) \ .\end{aligned}$$

So far, we have made exact computations. Now, we consider gravitational waves emitted by a non-relativistic isolated far-away source. We individuate the size of the source to be $r+\delta r$ which low frequency of emission $\delta r \ll 1/\omega$. Physically, it means that light travels faster than the components of the source. This means that

$$\frac{\exp(i\omega|\underline{x}-\underline{y}|)}{|\underline{x}-\underline{y}|} \simeq \frac{\exp(i\omega r)}{r} \ ,$$

hence

$$\tilde{h}^T_{\mu\nu}(\omega,\underline{x}) \simeq 4G_N \frac{\exp(i\omega r)}{r} \int d^3y \ \tilde{T}_{\mu\nu}(\omega,\underline{y}) \ .$$

Using the transverse gauge, we need only to find the spatial components. In fact, we have

$$\partial^\mu \tilde{h}_{\mu\nu}(t,\mathbf{x}) = 0 \ , \quad \tilde{h}^T_{0\nu} = -\frac{i}{\omega}\partial^i \tilde{h}_{i\nu} \ .$$

Let us restrict ourselves to the spatial components. Notice that

$$0 = \int d^3y \ \partial_k(y^i\tilde{T}^{kj}) = \int d^3y \ y^i\partial_k(\tilde{T}^{kj}) + \int d^3y \ \tilde{T}^{ij} \ ,$$

because it is a boundary term. Therefore, we have

$$\int d^3y \ \tilde{T}^{ij} = -\int d^3y \ y^i\partial_k(\tilde{T}^{kj}) \ . \tag{5.1}$$

Using the continuity equation, we find

$$\partial^\mu \tilde{T}_{\mu\nu} = 0 \ , \quad -\partial_k \tilde{T}^{k\nu} = i\omega T^{0\nu} \ .$$

Therefore,

$$\begin{aligned}\int d^3y \ T^{ij}(\omega,\underline{y}) &= -\int d^3y \ y^i\partial_k(\tilde{T}^{kj}) = i\omega \int d^3y \ y^i\tilde{T}^{0j} \\ &= \frac{i\omega}{2}\int d^3y \ (y^i\tilde{T}^{0j}+y^j\tilde{T}^{0i})\end{aligned}$$

where we have symmetrised, since T is symmetric. Exploiting again (5.1), we find

$$\int d^3y\ T^{ij}(\omega,\underline{y}) = \frac{i\omega}{2}\int d^3y\ (\partial_l(y^iy^l\tilde{T}^{0j}) - y^iy^j\partial_l\tilde{T}^{0l}) = -\frac{\omega^2}{2}\int d^3y\ y^iy^j\tilde{T}^{00}\ .$$

We define the quadrupole moment tensor of the energy density of the source

$$I^{ij}(t) = \int d^3y\ y^iy^jT^{00}\ .$$

Putting together, we obtain

$$\int d^3y\ T^{ij}(\omega,\underline{y}) = -\frac{\omega^2}{2}\int d^3y\ y^iy^j\tilde{T}^{00} = -\frac{\omega^2}{2}I^{ij}\ ,$$

$$\begin{aligned}\tilde{h}^T_{ij}(\omega,\underline{x}) &\simeq 4G_N\frac{\exp(i\omega r)}{r}\int d^3y\ \tilde{T}_{ij}(\omega,\underline{y})\\ &= -\frac{\omega^2}{2}\frac{\exp(i\omega r)}{r}4G_NI^{ij} = -2G_N\omega^2\frac{\exp(i\omega r)}{r}\tilde{I}_{ij}(\omega)\ .\end{aligned}$$

Finally, we find the quadrupole formula by an inverse Fourier transform

$$\begin{aligned}h^T_{ij}(t,\underline{x}) &= \int\frac{d\omega}{(2\pi)^{1/2}}\exp(-i\omega t)\tilde{h}^T_{ij}(\omega,\underline{x})\\ &= -2G_N\int\frac{d\omega}{(2\pi)^{1/2}}\exp(-i\omega t)\omega^2\frac{\exp(i\omega r)}{r}\tilde{I}_{ij}(\omega)\\ &= -\frac{2G_N}{r}\int\frac{d\omega}{(2\pi)^{1/2}}\omega^2\exp(-i\omega(t-r))\tilde{I}_{ij}\\ &= -\frac{2G_N}{r}\frac{d^2}{dt^2}I_{ij}(t_r)\ ,\end{aligned}$$

where we have used the Feynman technique and

$$I_{ij}(t_r) = \int\frac{d\omega}{(2\pi)^{1/2}}\exp(-i\omega t_r)I_{ij}(\omega)\ .$$

Physically, it means that the gravitational wave produced is proportional to the second derivative of the quadrupole moment of energy density.

5.4 Binary system

Consider the gravitational wave emitted by a binary system composed by two stars of equal mass M orbiting around their center of mass R in the plane $x^1 - x^2$. Using the Newtonian-Keplerian approximation, we can find the angular frequency of the orbit

$$\Omega = \left(\frac{G_NM}{4R^3}\right)^{1/2}\ .$$

Proof. Equating the centrigual force with the gravitational one

$$\frac{G_N M}{(2R)^2} = \frac{Mv^2}{R} \ , \quad v = \Big(\frac{G_N M}{4R}\Big)^{1/2} \ .$$

The frequency is given by

$$\Omega = \frac{2\pi}{T} = \frac{2\pi v}{2\pi R} = \Big(\frac{G_N M}{4R^3}\Big)^{1/2} \ .$$

q.e.d.

The explicit path is given by

$$x_a^1 = R\cos(\Omega t) \ , \quad x_a^2 = R\sin(\Omega t) \ , \quad x_b^1 = -R\cos(\Omega t) \ , \quad x_b^2 = -R\sin(\Omega t) \ ,$$

where a and b are different stars. The energy density is given by

$$T^{00}(t,\underline{x}) = M\delta(x^3)(\delta(x^1 - R\cos(\Omega t))\delta(x^2 - R\sin(\Omega t)) + \delta(x^1 + R\cos(\Omega t))\delta(x^2 + R\sin(\Omega t))) \ .$$

The quadrupole tensor is

$$I_{ij} = \begin{bmatrix} MR^2(1+\cos(2\Omega t)) & MR^2\sin(2\Omega t) & 0 \\ MR^2\sin(2\Omega t) & MR^2(1-\cos(2\Omega t)) & 0 \\ 0 & 0 & 0 \end{bmatrix} \ . \tag{5.2}$$

Proof. In fact

$$\begin{aligned} I_{11} &= \int d^3y \ T^{00}(t,\underline{y}) y_1 y_1 \\ &= \int d^3y \ M\delta^3(y^3)\Big(\delta(y^1 - R\cos(\Omega t))\delta(y^2 - R\sin(\Omega t)) \\ &\qquad + \delta(y^1 + R\cos(\Omega t))\delta(y^2 + R\sin(\Omega t))\Big) y_1 y_1 \\ &= M\Big(\int dy^1 \ y_1 y_1 \delta(y_1 - R\cos(\Omega t)) \int dy_2 \ \delta(y_2 - R\sin(\Omega t)) \int dy_3 \ \delta(y_3) \\ &\qquad + \int dy^1 \ y_1 y_1 \delta(y_1 + R\cos(\Omega t)) \int dy_2 \ \delta(y_2 + R\sin(\Omega t)) \int dy_3 \ \delta(y_3)\Big) \\ &= 2MR^2\cos^2(\Omega t) = MR^2(1+\cos(2\Omega t)) \ , \end{aligned}$$

$$
\begin{aligned}
I_{12} = I_{21} &= \int d^3y\ T^{00}(t,\underline{y})y_1y_2 \\
&= \int d^3y\ M\delta^3(y^3)\Big(\delta(y^1 - R\cos(\Omega t))\delta(y^2 - R\sin(\Omega t)) \\
&\qquad + \delta(y^1 + R\cos(\Omega t))\delta(y^2 + R\sin(\Omega t))\Big)y_1y_2 \\
&= M\Big(\int dy^1\ y_1\delta(y_1 - R\cos(\Omega t))\int dy_2\ y_2\delta(y_2 - R\sin(\Omega t))\int dy_3\ \delta(y_3) \\
&\qquad + \int dy^1\ y_1\delta(y_1 + R\cos(\Omega t))\int dy_2\ y_2\delta(y_2 + R\sin(\Omega t))\int dy_3\ \delta(y_3)\Big) \\
&= 2MR^2\sin(\Omega t)\cos(\Omega t) = MR^2\sin(2\Omega t)\ ,
\end{aligned}
$$

$$
\begin{aligned}
I_{22} &= \int d^3y\ T^{00}(t,\underline{y})y_2y_2 \\
&= \int d^3y\ M\delta^3(y^3)\Big(\delta(y^1 - R\cos(\Omega t))\delta(y^2 - R\sin(\Omega t)) \\
&\qquad + \delta(y^1 + R\cos(\Omega t))\delta(y^2 + R\sin(\Omega t))\Big)y_2y_2 \\
&= M\Big(\int dy^1\ \delta(y_1 - R\cos(\Omega t))\int dy_2\ y_2y_2\delta(y_2 - R\sin(\Omega t))\int dy_3\ \delta(y_3) \\
&\qquad + \int dy^1\ \delta(y_1 + R\cos(\Omega t))\int dy_2\ y_2y_2\delta(y_2 + R\sin(\Omega t))\int dy_3\ \delta(y_3)\Big) \\
&= 2MR^2\sin^2(\Omega t) = MR^2(1 - \cos(2\Omega t))\ .
\end{aligned}
$$

q.e.d.

The gravitational wave energy-momentum tensor in the transverse traceless gauge is given by

$$t_{\mu\nu} = \frac{1}{32\pi G_N}\langle\partial_\mu h^{TT\alpha\beta}\partial_\nu h^{TT}_{\alpha\beta}\rangle\ .$$

Since it is in the transverse traceless gauge, we need to introduce the reduced quadrupole moment

$$I^{TT}_{ij} = I_{ij} - \frac{1}{3}\delta_{ij}\delta^{kl}I_{kl}\ ,$$

and the quadrupole formula becomes

$$h^{TT}_{ij} = \frac{2G_N}{r}\frac{d^2}{dt^2}I^{TT}_{ij}(t_r)\ .$$

The power emitted by the system is given by

$$P = -\frac{G_N}{5}\langle\frac{d^3}{dt^3}I^{TT}_{ij}\frac{d^3}{dt^3}I^{TTij}\rangle\ .$$

In our binary merge case, we have the reduced quadrupole moment to be

$$I_{ij}^{TT} = \frac{MR^2}{3}\begin{bmatrix} 1+3\cos(2\Omega t) & 3\sin(2\Omega t) & 0 \\ 3\sin(2\Omega t) & 1-3\cos(2\Omega t) & 0 \\ 0 & 0 & -2 \end{bmatrix} .$$

Proof. In fact, the trace of (5.2) is

$$\delta^{ij} I_{ij} = 2MR^2 .$$

The reduced quadrupole moment is

$$\begin{aligned} I_{ij}^{TT} &= I_{ij} - \frac{1}{3}\delta_{ij}\delta^{kl} I_{kl} \\ &= \begin{bmatrix} MR^2(1+\cos(2\Omega t)) & MR^2\sin(2\Omega t) & 0 \\ MR^2\sin(2\Omega t) & MR^2(1-\cos(2\Omega t)) & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{2}{3}MR^2 \\ &= \frac{MR^2}{3}\begin{bmatrix} 1+3\cos(2\Omega t) & 3\sin(2\Omega t) & 0 \\ 3\sin(2\Omega t) & 1-3\cos(2\Omega t) & 0 \\ 0 & 0 & -2 \end{bmatrix} \end{aligned}$$

q.e.d.

5.5 BOH

We can approximate on the center of mass $\underline{r}$, so that $h_{ij}^T(t,r)$. Outside the source, we can find the solution using the Green function of the d'Alambertian

$$G_{ijkl}^T(t,r,t',r') \sim \frac{\delta(t'-t+r)}{r} ,$$

hence

$$h_{ij}^T(t,r) = 16\pi G_N \int d^3r' \; G_{ijkl}^T(t,r,t',r') T^{kl}(t',r') ,$$

such that $G_{ijkl} = G_{jikl}$ and $G^i{}_{ikl} = 0$. Noticing that

$$T^{ij}(t',r) \simeq \rho(t',r')\delta\dot{r}^i(t')\dot{r}^j(t') ,$$

$$\int d^3r' \; T^{ij}(t',r') \simeq \int d^3x \; \rho(t',\underline{r}+\underline{x})\dot{x}^i\dot{x}^j \simeq \frac{d^2}{dt^2}\int d^3x \; \rho(t',\underline{r}+\underline{x})x^i x^j ,$$

where we have neglected $\dot{\rho}$ and $\ddot{x}^i$, we define the traceless quadrupole

$$I_{ij} \simeq \int d^3x \; \rho(\underline{r}+\underline{x})\Big(x^i x^j - \frac{1}{3}\delta_{ij}x^2\Big)$$

and we find the quadrupole formula

$$h_{ij}^T(t,r) \simeq \frac{16\pi G_N}{r} \ddot{I}_{ij}(t-r) \ ,$$

where $t-r$ is the retarded.

Since gravity is described by the metric, there is no force and therefore no energy. However, we can identify the energy of a gravitational wave by the work done on matter. In our approximation, this is

$$\dot{E} \sim G_N \dddot{I}_{ij} \dddot{I}^{ij} \ .$$

Consider a binary system of orbital radius R and velocity

$$v^2 \sim \frac{G_N M}{R} \ .$$

The strain is

$$a \sim \frac{\Delta l}{l} \sim \frac{G_N M v^2}{r} \sim \frac{G_N^2 M^2}{rR}$$

Furthermore, the frequency is

$$\omega \sim \frac{v}{R} \sim \sqrt{\frac{G_N M}{R^3}} \ .$$

5.6 Detection

We can detect gravitational waves by means of a Michelson-Morley interferometer. Assuming $v \ll c$ and $t \sim l_0 \ll l_0/v$, we find the maximum travel time

$$\Delta t \simeq 2(l_0(1+a)^{1/2} - l_0(1-a)^{1/2}) \simeq a l_0 \ .$$

Looking at the interference pattern, we find that it is proportional to the strain

$$a \sim \frac{\Delta l}{l} \simeq \frac{\Delta l}{l} \ .$$

Part III

Black holes

Chapter 6

Exterior Schwarzschild metric

In this chapter, we will solve the Einstein field equation for a symmetric static solution and will find the exterior Schwarzschild metric.

6.1 Schwarzschild metric

Consider a spherical symmetric source. We are interested in the outside region, where $T^{\mu\nu} = 0$. The Einstein field equations become

$$G_{\mu\nu} = 0 \ .$$

It is possible to prove that they correspond to

$$R_{\mu\nu} = 0 \ .$$

Proof. In fact, we take the trace

$$g^{\mu\nu} G_{\mu\nu} = g^{\mu\nu}(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R) = R - 2R = 0 \ ,$$

hence,

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \underbrace{R}_{0} = R_{\mu\nu} \ .$$

q.e.d.

We can exploit the symmetries to simplify the metric. We assumed a static metric, which means that there exists a time-like Killing vector

$$\underline{K}_t = \frac{\partial}{\partial t} \ .$$

We also assumed spherical symmetry, which means that there exist three space-like Killing vectors

$$\underline{K}_i = \frac{d}{d\theta_i} \ ,$$

where $i = 1, 2, 3$. These spatial vectors must be conserved in time, therefore

$$[\underline{K}_t, \underline{K}_i] = 0 \ ,$$

and we can write down the metric with temporal part orthogonal to the spatial one with the latter in polar coordinates

$$g_{\mu\nu} = \begin{bmatrix} -A(r) & 0 & 0 & 0 \\ 0 & B(r) & 0 & 0 \\ 0 & 0 & C(r) & 0 \\ 0 & 0 & 0 & C(r)\sin^2\theta \end{bmatrix} = \begin{bmatrix} -A(r) & 0 & 0 & 0 \\ 0 & B(r) & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{bmatrix} \ ,$$

$$ds^2 = -A(r)dt^2 + B(r)dr^2 + C(r)(d\theta^2 + \sin^2\theta d\phi^2) = -A(r)dt^2 + B(r)dr^2 + r^2 d\Omega^2 \ ,$$

where we have used the rescaling freedom to set $C(r) = r^2$. In this way, r becomes the areal radius, since the area of a sphere is given by

$$S(r) = \int d\theta^2 = r^2 \int_0^\pi d\theta \sin\theta \int_0^{2\pi} d\phi = 4\pi r^2 \ .$$

Notice that the proper radius is instead

$$R(r) = \int_0^r dx \ \sqrt{g_{rr}} = \int_0^r dx \ \sqrt{B} \ . \tag{6.1}$$

In order to find a chart, we need to take into account also the domain of the coordinates. We reasonably set $t \in \mathbb{R}$, $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi[$. However, we cannot say anything about r. We need to find the manifold, not only the metric. In fact, it can be possible that this chart does not cover the entire manifold.

6.2 Computations

Now, we compute the Einstein's field equations. The metric is

$$g_{\mu\nu} = \begin{bmatrix} A & 0 & 0 & 0 \\ 0 & B & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{bmatrix} \ ,$$

whereas its inverse is

$$g^{\mu\nu} = \begin{bmatrix} \frac{1}{A} & 0 & 0 & 0 \\ 0 & \frac{1}{B} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2\sin^2\theta} \end{bmatrix} \ .$$

The derivative with respect to r is

$$g_{\mu\nu,1} = \begin{bmatrix} A' & 0 & 0 & 0 \\ 0 & B' & 0 & 0 \\ 0 & 0 & 2r & 0 \\ 0 & 0 & 0 & 2r\sin^2\theta \end{bmatrix} ,$$

whereas its inverse is

$$g_{\mu\nu,2} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2r^2\sin\theta\cos\theta \end{bmatrix} .$$

The Christoffel symbols induced by the metric are

$$\Gamma^k_{ij} = \frac{1}{2} g^{kl}(g_{il,j} + g_{jl,i} - g_{ij,l}) .$$

Proof. The 0-th Christoffel symbols are

$$\Gamma^0_{00} = \frac{1}{2} g^{0l}(g_{0l,0} + g_{0l,0} - g_{00,l}) = \frac{1}{2} g^{00}(g_{00,0} + g_{00,0} - g_{00,0}) = 0 ,$$

$$\Gamma^0_{11} = \frac{1}{2} g^{0l}(g_{1l,1} + g_{1l,1} - g_{11,l}) = \frac{1}{2} g^{00}(g_{10,1} + g_{10,1} - g_{11,0}) = 0 ,$$

$$\Gamma^0_{22} = \frac{1}{2} g^{0l}(g_{2l,2} + g_{2l,2} - g_{22,l}) = \frac{1}{2} g^{00}(g_{20,2} + g_{20,2} - g_{22,0}) = 0 ,$$

$$\Gamma^0_{33} = \frac{1}{2} g^{0l}(g_{3l,3} + g_{3l,3} - g_{33,l}) = \frac{1}{2} g^{00}(g_{30,3} + g_{30,3} - g_{33,0}) = 0 ,$$

$$\Gamma^0_{01} = \Gamma^0_{10} = \frac{1}{2} g^{0l}(g_{1l,0} + g_{0l,1} - g_{10,l}) = \frac{1}{2} \underbrace{g^{00}}_{\frac{1}{A}} (g_{10,0} + \underbrace{g_{00,1}}_{A'} - g_{10,0}) = \frac{A'}{2A} ,$$

$$\Gamma^0_{02} = \Gamma^0_{20} = \frac{1}{2} g^{0l}(g_{2l,0} + g_{0l,2} - g_{20,l}) = \frac{1}{2} g^{00}(g_{20,0} + g_{00,2} - g_{20,0}) = 0 ,$$

$$\Gamma^0_{03} = \Gamma^0_{30} = \frac{1}{2} g^{0l}(g_{3l,0} + g_{0l,3} - g_{30,l}) = \frac{1}{2} g^{00}(g_{30,0} + g_{00,3} - g_{30,0}) = 0 ,$$

$$\Gamma^0_{12} = \Gamma^0_{21} = \frac{1}{2} g^{0l}(g_{2l,1} + g_{1l,2} - g_{21,l}) = \frac{1}{2} g^{00}(g_{20,1} + g_{10,2} - g_{21,0}) = 0 ,$$

$$\Gamma^0_{13} = \Gamma^0_{31} = \frac{1}{2} g^{0l}(g_{3l,1} + g_{1l,3} - g_{31,l}) = \frac{1}{2} g^{00}(g_{30,1} + g_{10,3} - g_{31,0}) ,$$

$$\Gamma^0_{23} = \Gamma^0_{32} = \frac{1}{2} g^{0l}(g_{3l,2} + g_{2l,3} - g_{32,l}) = \frac{1}{2} g^{00}(g_{30,2} + g_{20,3} - g_{32,0}) = 0 .$$

The 1-st Christoffel symbols are

$$\Gamma^1_{00} = \frac{1}{2} g^{1l}(g_{0l,0} + g_{0l,0} - g_{00,l}) = \frac{1}{2} \underbrace{g^{11}}_{\frac{1}{B}} (g_{01,0} + g_{01,0} - \underbrace{g_{00,1}}_{A'}) = -\frac{A'}{2B} \ ,$$

$$\Gamma^1_{11} = \frac{1}{2} g^{1l}(g_{1l,1} + g_{1l,1} - g_{11,l}) = \frac{1}{2} \underbrace{g^{11}}_{\frac{1}{B}} (\underbrace{g_{11,1}}_{B'} + g_{11,1} - g_{11,1}) = \frac{B'}{2B} \ ,$$

$$\Gamma^1_{22} = \frac{1}{2} g^{1l}(g_{2l,2} + g_{2l,2} - g_{22,l}) = \frac{1}{2} \underbrace{g^{11}}_{\frac{1}{B}} (g_{21,2} + g_{21,2} - \underbrace{g_{22,1}}_{2r}) = -\frac{r}{B} \ ,$$

$$\Gamma^1_{33} = \frac{1}{2} g^{1l}(g_{3l,3} + g_{3l,3} - g_{33,l}) = \frac{1}{2} \underbrace{g^{11}}_{\frac{1}{B}} (g_{31,3} + g_{31,3} - \underbrace{g_{33,1}}_{2r\sin^2\theta}) = -\frac{r\sin^2\theta}{B} \ ,$$

$$\Gamma^1_{01} = \Gamma^1_{10} = \frac{1}{2} g^{1l}(g_{1l,0} + g_{0l,1} - g_{10,l}) = \frac{1}{2} g^{11}(g_{11,0} + g_{01,1} - g_{10,1}) = 0 \ ,$$

$$\Gamma^1_{02} = \Gamma^1_{20} = \frac{1}{2} g^{1l}(g_{2l,0} + g_{0l,2} - g_{20,l}) = \frac{1}{2} g^{11}(g_{21,0} + g_{01,2} - g_{20,1}) = 0 \ ,$$

$$\Gamma^1_{03} = \Gamma^1_{30} = \frac{1}{2} g^{1l}(g_{3l,0} + g_{0l,3} - g_{30,l}) = \frac{1}{2} g^{11}(g_{31,0} + g_{01,3} - g_{30,1}) = 0 \ ,$$

$$\Gamma^1_{12} = \Gamma^1_{21} = \frac{1}{2} g^{1l}(g_{2l,1} + g_{1l,2} - g_{21,l}) = \frac{1}{2} g^{11}(g_{21,1} + g_{11,2} - g_{21,1}) = 0 \ ,$$

$$\Gamma^1_{13} = \Gamma^1_{31} = \frac{1}{2} g^{1l}(g_{3l,1} + g_{1l,3} - g_{31,l}) = \frac{1}{2} g^{11}(g_{31,1} + g_{11,3} - g_{31,1}) = 0 \ ,$$

$$\Gamma^1_{23} = \Gamma^1_{32} = \frac{1}{2} g^{1l}(g_{3l,2} + g_{2l,3} - g_{32,l}) = \frac{1}{2} g^{11}(g_{31,2} + g_{21,3} - g_{32,1}) = 0 \ .$$

The 2-nd Christoffel symbols are

$$\Gamma^2_{00} = \frac{1}{2} g^{2l}(g_{0l,0} + g_{0l,0} - g_{00,l}) = \frac{1}{2} g^{22}(g_{02,0} + g_{02,0} - g_{00,2}) = 0 \ ,$$

$$\Gamma^2_{11} = \frac{1}{2} g^{2l}(g_{1l,1} + g_{1l,1} - g_{11,l}) = \frac{1}{2} g^{22}(g_{12,1} + g_{12,1} - g_{11,2}) = 0 \ ,$$

$$\Gamma^2_{22} = \frac{1}{2} g^{2l}(g_{2l,2} + g_{2l,2} - g_{22,l}) = \frac{1}{2} g^{22}(g_{22,2} + g_{22,2} - g_{22,2}) = 0 \ ,$$

$$\Gamma^2_{33} = \frac{1}{2} g^{2l}(g_{3l,3} + g_{3l,3} - g_{33,l}) = \frac{1}{2} \underbrace{g^{22}}_{\frac{1}{r^2}} (g_{32,3} + g_{32,3} - \underbrace{g_{33,2}}_{2r^2\sin\theta\cos\theta}) = -\sin\theta\cos\theta \ ,$$

$$\Gamma^2_{01} = \Gamma^2_{10} = \frac{1}{2} g^{2l}(g_{1l,0} + g_{0l,1} - g_{10,l}) = \frac{1}{2} g^{22}(g_{12,0} + g_{02,1} - g_{10,2}) = 0 \ ,$$

$$\Gamma^2_{02} = \Gamma^2_{20} = \frac{1}{2} g^{2l}(g_{2l,0} + g_{0l,2} - g_{20,l}) = \frac{1}{2} g^{22}(g_{22,0} + g_{02,2} - g_{20,2}) = 0 \ ,$$

$$\Gamma^2_{03} = \Gamma^2_{30} = \frac{1}{2} g^{2l}(g_{3l,0} + g_{0l,3} - g_{30,l}) = \frac{1}{2} g^{22}(g_{32,0} + g_{02,3} - g_{30,2}) = 0 \ ,$$

$$\Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{2} g^{2l}(g_{2l,1} + g_{1l,2} - g_{21,l}) = \frac{1}{2} \underbrace{g^{22}}_{\frac{1}{r^2}} (\underbrace{g_{22,1}}_{2r} + g_{12,2} - g_{21,2}) = \frac{1}{r} \ ,$$

$$\Gamma^2_{13} = \Gamma^2_{31} = \frac{1}{2} g^{2l}(g_{3l,1} + g_{1l,3} - g_{31,l}) = \frac{1}{2} g^{22}(g_{32,1} + g_{12,3} - g_{31,2}) = 0 \ ,$$

$$\Gamma^2_{23} = \Gamma^2_{32} = \frac{1}{2} g^{2l}(g_{3l,2} + g_{2l,3} - g_{32,l}) = \frac{1}{2} g^{22}(g_{32,2} + g_{22,3} - g_{32,2}) = 0 \ .$$

The 3-rd Christoffel symbols are

$$\Gamma^3_{00} = \frac{1}{2} g^{3l}(g_{0l,0} + g_{0l,0} - g_{00,l}) = \frac{1}{2} g^{33}(g_{03,0} + g_{03,0} - g_{00,3}) = 0 \ ,$$

$$\Gamma^3_{11} = \frac{1}{2} g^{3l}(g_{1l,1} + g_{1l,1} - g_{11,l}) = \frac{1}{2} g^{33}(g_{13,1} + g_{13,1} - g_{11,3}) = 0 \ ,$$

$$\Gamma^3_{22} = \frac{1}{2} g^{3l}(g_{2l,2} + g_{2l,2} - g_{22,l}) = \frac{1}{2} g^{33}(g_{23,2} + g_{23,2} - g_{22,3}) = 0 \ ,$$

$$\Gamma^3_{33} = \frac{1}{2} g^{3l}(g_{3l,3} + g_{3l,3} - g_{33,l}) = \frac{1}{2} g^{33}(g_{33,3} + g_{33,3} - g_{33,3}) = 0 \ ,$$

$$\Gamma^3_{01} = \Gamma^3_{10} = \frac{1}{2} g^{3l}(g_{1l,0} + g_{0l,1} - g_{10,l}) = \frac{1}{2} g^{33}(g_{13,0} + g_{03,1} - g_{10,3}) = 0 \ ,$$

$$\Gamma^3_{02} = \Gamma^3_{20} = \frac{1}{2} g^{3l}(g_{2l,0} + g_{0l,2} - g_{20,l}) = \frac{1}{2} g^{33}(g_{23,0} + g_{03,2} - g_{20,3}) = 0 \ ,$$

$$\Gamma^3_{03} = \Gamma^3_{30} = \frac{1}{2} g^{3l}(g_{3l,0} + g_{0l,3} - g_{30,l}) = \frac{1}{2} g^{33}(g_{33,0} + g_{03,3} - g_{30,3}) = 0 \ ,$$

$$\Gamma^3_{12} = \Gamma^3_{21} = \frac{1}{2} g^{3l}(g_{2l,1} + g_{1l,2} - g_{21,l}) = \frac{1}{2} g^{33}(g_{23,1} + g_{13,2} - g_{21,3}) = 0 \ ,$$

$$\Gamma^3_{13} = \Gamma^3_{31} = \frac{1}{2} g^{3l}(g_{3l,1} + g_{1l,3} - g_{31,l}) = \frac{1}{2} \underbrace{g^{33}}_{\frac{1}{r^2 \sin^2\theta}} (\underbrace{g_{33,1}}_{2r\sin^2\theta} + g_{13,3} - g_{31,3}) = \frac{1}{r} \ ,$$

$$\Gamma^3_{23} = \Gamma^3_{32} = \frac{1}{2} g^{3l}(g_{3l,2} + g_{2l,3} - g_{32,l}) = \frac{1}{2} \underbrace{g^{33}}_{\frac{1}{r^2 \sin^2\theta}} (\underbrace{g_{33,2}}_{2r^2\sin\theta\cos\theta} + g_{23,3} - g_{32,3}) = \frac{\cos\theta}{\sin\theta} \ .$$

Hence, the non-zero Christoffel symbols are

$$\Gamma^0_{01} = \Gamma^0_{10} = \frac{A'}{2A} \ , \quad \Gamma^1_{00} = -\frac{A'}{2B} \ , \quad \Gamma^1_{11} = \frac{B'}{2B} \ , \quad \Gamma^1_{22} = -\frac{r}{B} \ , \quad \Gamma^1_{33} = -\frac{r\sin^2\theta}{B} \ ,$$

$$\Gamma^2_{33} = -\sin\theta\cos\theta \ , \quad \Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{r} \ , \quad \Gamma^3_{13} = \Gamma^3_{31} = \frac{1}{r} \ , \quad \Gamma^3_{23} = -\frac{\cos\theta}{\sin\theta} \ .$$

The derivatives are

$$\Gamma^0_{01,1} = \Gamma^0_{10,1} = \frac{\partial}{\partial r}\Big(\frac{A'}{2A}\Big) = \frac{A''A - (A')^2}{2A^2} \ ,$$

$$\Gamma^1_{00,1} = \frac{\partial}{\partial r}\Big(-\frac{A'}{2B}\Big) = -\frac{A''B - A'B'}{2B^2} \ ,$$

$$\Gamma^1_{11,1} = \frac{\partial}{\partial r}\Big(\frac{B'}{2B}\Big) = \frac{B''B - (B')^2}{2B^2} \ ,$$

$$\Gamma^1_{22,1} = \frac{\partial}{\partial r}\Big(-\frac{r}{B}\Big) = -\frac{B - rB'}{B^2} \ ,$$

$$\Gamma^1_{33,1} = \frac{\partial}{\partial r}\Big(-\frac{r\sin^2\theta}{B}\Big) = -\sin^2\theta\frac{B - rB'}{B^2} \ ,$$

$$\Gamma^1_{33,2} = \frac{\partial}{\partial \theta}\Big(-\frac{r\sin^2\theta}{B}\Big) = -\frac{2r\sin\theta\cos\theta}{B^2} \ ,$$

$$\Gamma^2_{33,2} = \frac{\partial}{\partial \theta}(\sin\theta\cos\theta) = \cos^2\theta - \sin^2\theta \ ,$$

$$\Gamma^2_{12,1} = \Gamma^2_{21,1} = \frac{\partial}{\partial r}\Big(\frac{1}{r}\Big) = -\frac{1}{r^2} \ ,$$

$$\Gamma^3_{13,1} = \Gamma^3_{31,1} = \frac{\partial}{\partial r}\Big(\frac{1}{r}\Big) = -\frac{1}{r^2} \ ,$$

$$\Gamma^3_{23,2} = \frac{\partial}{\partial \theta}\cot\theta = -\frac{1}{\sin^2\theta} \ .$$

The Ricci tensor is

$$R_{ij} = \Gamma^k_{ij,k} - \Gamma^k_{ik,j} + \Gamma^a_{ij}\Gamma^k_{ak} - \Gamma^a_{ik}\Gamma^k_{aj} \ .$$

Its components are

$$\begin{aligned}
R_{00} &= \Gamma^k_{00,k} - \Gamma^k_{0k,0} + \Gamma^a_{00}\Gamma^k_{ak} - \Gamma^a_{0k}\Gamma^k_{a0} \\
&= \Gamma^1_{00,1} + \Gamma^1_{00}\Gamma^0_{10} + \Gamma^1_{00}\Gamma^1_{11} + \Gamma^1_{00}\Gamma^2_{12} + \Gamma^1_{00}\Gamma^3_{13} - \Gamma^0_{01}\Gamma^1_{00} - \Gamma^1_{00}\Gamma^0_{10} \\
&= -\frac{A''B - A'B'}{2B^2} - \cancel{\frac{A'}{2B}\frac{A'}{2A}} - \frac{A'}{2B}\frac{B'}{2B} - \frac{A'}{2rB} - \frac{A'}{2rB} + \frac{A'}{2A}\frac{A'}{2B} + \cancel{\frac{A'}{2B}\frac{A'}{2A}} \\
&= -\frac{A''}{2B} + \frac{A'B'}{4B^2} - \frac{A'}{rB} + \frac{(A')^2}{4AB} \ ,
\end{aligned}$$

$$\begin{aligned}
R_{11} &= \Gamma^k_{11,k} - \Gamma^k_{1k,1} + \Gamma^a_{11}\Gamma^k_{ak} - \Gamma^a_{1k}\Gamma^k_{a1} \\
&= \cancel{\Gamma^1_{11,1}} - \Gamma^0_{10,1} - \cancel{\Gamma^1_{11,1}} - \Gamma^2_{12,1} - \Gamma^3_{13,1} + \Gamma^1_{11}\Gamma^0_{10} + \cancel{\Gamma^1_{11}\Gamma^1_{11}} \\
&\quad + \Gamma^1_{11}\Gamma^2_{12} + \Gamma^1_{11}\Gamma^3_{13} - \Gamma^0_{10}\Gamma^0_{01} - \cancel{\Gamma^1_{11}\Gamma^1_{11}} - \Gamma^2_{12}\Gamma^2_{21} - \Gamma^3_{13}\Gamma^3_{31} \\
&= -\frac{A''A - (A'^2)}{2A^2} + \cancel{\frac{1}{r^2}} + \cancel{\frac{1}{r^2}} + \frac{B'}{2B}\frac{A'}{2A} + \frac{B'}{2rB} + \frac{B'}{2rB} - \frac{(A')^2}{4A^2} - \cancel{\frac{1}{r^2}} - \cancel{\frac{1}{r^2}} \\
&= -\frac{A''}{2A} + \frac{(A')^2}{4A^2} + \frac{B'A'}{4AB} + \frac{B'}{rB} \ ,
\end{aligned}$$

$$
\begin{aligned}
R_{22} &= \Gamma^k_{22,k} - \Gamma^k_{2k,2} + \Gamma^a_{22}\Gamma^k_{ak} - \Gamma^a_{2k}\Gamma^k_{a2} \\
&= \Gamma^1_{22,1} - \Gamma^3_{23,2} + \Gamma^1_{22}\Gamma^0_{10} + \Gamma^1_{22}\Gamma^1_{11} + \cancel{\Gamma^1_{22}\Gamma^2_{12}} \\
&\quad + \Gamma^1_{22}\Gamma^3_{13} - \cancel{\Gamma^1_{22}\Gamma^2_{12}} - \Gamma^2_{21}\Gamma^1_{22} - \Gamma^3_{23}\Gamma^3_{32} \\
&= -\frac{B - rB'}{B^2} + \frac{1}{\sin^2\theta} - \frac{r}{B}\frac{A'}{2A} - \frac{r}{B}\frac{B'}{2B} - \cancel{\frac{r}{B}}\frac{\cancel{1}}{r} + \frac{1}{\cancel{r}}\frac{\cancel{r}}{B} - \cot^2\theta \\
&= 1 - \frac{1}{B} + \frac{rB'}{2B^2} - \frac{rA'}{2AB} \ ,
\end{aligned}
$$

$$
\begin{aligned}
R_{33} &= \Gamma^k_{33,k} - \Gamma^k_{3k,3} + \Gamma^a_{33}\Gamma^k_{ak} - \Gamma^a_{3k}\Gamma^k_{a3} \\
&= \Gamma^1_{33,1} + \Gamma^2_{33,2} + \Gamma^1_{33}\Gamma^0_{10} + \Gamma^1_{33}\Gamma^1_{11} + \Gamma^1_{33}\Gamma^2_{12} + \cancel{\Gamma^2_{33}\Gamma^3_{23}} + \cancel{\Gamma^1_{33}\Gamma^3_{13}} - \cancel{\Gamma^1_{33}\Gamma^3_{13}} - \cancel{\Gamma^2_{33}\Gamma^3_{23}} - \Gamma^3_{31}\Gamma^1_{33} - \Gamma^3_{32}\Gamma^2_{3} \\
&= -\sin^2\theta\frac{B - rB'}{B^2} + \cancel{\cos^2\theta} - \sin^2\theta - \frac{r\sin^2\theta}{B}\frac{A'}{2A} - \frac{r\sin^2\theta}{B}\frac{B'}{2B} - \cancel{\frac{r\sin^2\theta}{B}}\frac{1}{r} \\
&\quad - \frac{1}{\cancel{r}}\cancel{\frac{r\sin^2\theta}{B}} - \cancel{\frac{\cos\theta}{\sin\theta}\sin\theta\cos\theta} \\
&= \sin^2\theta - \sin^2\theta\frac{1}{B} + \sin^2\theta\frac{rB'}{2B^2} - \sin^2\theta\frac{rA'}{2AB} \\
&= \sin^2\theta R_{22} \ .
\end{aligned}
$$

Finally, we find a system of differential equations

$$
\begin{aligned}
R_{00} &= -\frac{A''}{2B} + \frac{(A')^2}{4AB} + \frac{A'B'}{4B^2} - \frac{A'}{rB} = 0 \ , \\
R_{11} &= -\frac{A''}{2A} + \frac{(A')^2}{4A^2} + \frac{B'A'}{4AB} + \frac{B'}{rB} = 0 \ , \\
R_{22} &= 1 - \frac{1}{B} + \frac{rB'}{2B^2} - \frac{rA'}{2AB} = 0 \ , \\
R_{33} &= \sin^2\theta R_{22} = 0 \ .
\end{aligned}
$$

We notice that

$$
\begin{aligned}
0 &= \frac{B}{A}R_{00} - R_{11} \\
&= -\cancel{\frac{B}{A}\frac{A''}{2B}} + \cancel{\frac{B}{A}\frac{(A')^2}{4AB}} + \cancel{\frac{B}{A}\frac{A'B'}{4B^2}} - \frac{B}{A}\frac{A'}{rB} + \cancel{\frac{A''}{2A}} - \cancel{\frac{(A')^2}{4A^2}} - \cancel{\frac{B'A'}{4AB}} - \frac{B'}{rB} \\
&= -\frac{A'}{rA} - \frac{B'}{rB} \ .
\end{aligned}
$$

Hence

$$
0 = A'B + B'A = (AB)' \ ,
$$

which implies, after a time rescaling, that

$$
A = -B^{-1} \ .
$$

Therefore, the third equation becomes

$$0 = 1 - \frac{1}{B} + \frac{rB'}{2B^2} - \frac{rA'}{2AB} = 1 + A - rA, \ ,$$

or, equivalently,

$$(rA)' = 1 \ .$$

Its solution is

$$A = 1 + \frac{2K}{r} \ .$$

q.e.d.

The constant K can be found by considering the Newtonian limit and it is

$$K = G_N M \ .$$

Proof. In fact, by looking at the Newton equation

$$\frac{d^2x^i}{dt^2} = \frac{1}{2}h_{00,i}$$

and comparing it to

$$A = g_{00} = 1 + \frac{2K}{r} = \eta_{00} + h_{00} \ ,$$

we find

$$h_{00} = \frac{2K}{r} = -2V_N = \frac{2G_N M}{r} \ ,$$

where V_N is the gravitational potential. Hence,

$$K = G_N M \ .$$

q.e.d.

At last, we have obtained the exterior Schwarzschild metric

$$g_{\mu\nu} = \begin{bmatrix} -(1-\frac{2G_N M}{r}) & 0 & 0 & 0 \\ 0 & (1-\frac{2G_N M}{r})^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{bmatrix} \ ,$$

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu = -\Big(1 - \frac{2G_N M}{r}\Big)dt^2 + \Big(1 - \frac{2G_N M}{r}\Big)^{-1}dr^2 + r^2 d\theta^2 + r^2\sin^2\theta d\phi^2 \ .$$

The quantity $R_H = 2G_N M$ is called the Schwarzschild radius. Notice that at spatial infinity, the metric becomes the Minkowski one, a property called the Minkowski flatness.

There is a theorem, due to Birkhoff, that helps us understand why we do not have $A(t, r)$.

Theorem 6.1 (Birkhoff)
A spherically symmetric metric which solves the vacuum Einstein field equations is static.

6.3 Areal radius

Static observes are inertial at spatial infinity $r \gg R_H$, and they differ from it the more they approach R_H. The proper radius of a sphere (**??**) is

$$R(r) = r\sqrt{1-\frac{r_H}{r}} + \frac{r_H}{2}\ln\Big(\frac{2r}{r_H}\Big(1+\sqrt{1-\frac{r_H}{r}}-1\Big)\Big) \ .$$

Proof. Maybe in the future. q.e.d.

6.4 Radial geodesics

Since the Schwarzschild metric does not carry dependence of the source size, we can suppose that $r_S \ll R_H$ and the metric becomes for $r \to R_H$, $g_{tt} = 0$ and $g_{rr} \to \infty$. Furthermore, the signs of g_{tt} and g_{rr} change across $r = R_H$. The Killing vector $\underline{K}_t$ is defined everywhere but $\underline{K}_r$ is not defined. This means that for inside and outside R_H, we maybe need two different manifolds, with the same coordinates (t, r, θ, ϕ) but different domains $r \in (0, R_H)$ for inside and $r \in (R_H, \infty)$ for outside. Notice that we need to find a chart that covers R_H. Therefore, we need to study a test particle that travels from outside to inside.

Consider the action of an infalling massive test particle

$$S = m\int_0^\tau d\tau\ \sqrt{-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu} = m\int_0^\tau d\tau\ \sqrt{2T} \ ,$$

where

$$2T = (1-\frac{2G_N M}{r})\dot{t}^2 - (1-\frac{2G_N M}{r})^{-1}\dot{r}^2 - r^2\dot{\theta}^2 - r^2\sin^2\theta\dot{\phi}^2 = 1 \ .$$

The Euler-Lagrange equations are

$$\frac{d}{d\tau}\Big(\frac{\partial T}{\partial \dot{x}^\mu}\Big) - \frac{\partial T}{\partial x^\mu} = 0 \ ,$$

because

$$\delta S = m\delta\int_0^\tau d\tau\ \sqrt{2T} = m\int_0^\tau d\tau\frac{\delta T}{\sqrt{2T}} = m\delta\int_0^\tau d\tau\ T \ .$$

Since the Lagrangian does not depend on t and ϕ, we can define two conserved quantities: energy

$$E = \frac{\partial T}{\partial \dot{t}} = (1 - \frac{2G_N M}{r})\dot{t} \ ,$$

and angular momentum

$$L = \frac{\partial T}{\partial \dot{\phi}} = -r^2 \sin^2\theta \dot{\phi} \ .$$

Notice that the energy reduces to 1 in the weak-field limit. By means of the Killing vectors $\underline{K}_i$, we can rotate the reference frame in order to have $\theta = \pi/2$. In fact, the Euler-Lagrange equation for θ are identically satisfied

$$0 = \frac{d}{d\tau}\Big(\frac{\partial T}{\partial \dot{\theta}}\Big) - \frac{\partial T}{\partial \theta} = \ddot{\theta} - \sin\theta\cos\theta\dot{\phi}^2 \ .$$

Now, consider the simple case $\phi = 0$ of radially in-falling particles. The energy equation becomes

$$2T = (1 - \frac{2G_N M}{r}\Big)^{-1}(E^2 - \dot{r}^2) = 1 \ ,$$

where we have set $\theta = \pi/2$ and $\phi = 0$. We solve this equation and we find

$$\dot{t} = (1 - \frac{2G_N M}{r}\Big)^{-1} E \ ,$$

which implies that

$$\frac{dt}{d\tau} \sim \frac{1}{r - R_H} \ .$$

Physically, it means that an asymptotically inertial observer at spatial infinity will take an infinite amount of coordinate time to see a particle falling radially. Now, we invert the energy equation

$$\dot{r}^2 = \frac{2G_N M}{r} + (E^2 - 1) \ ,$$

and we derive it

$$\ddot{r} = -\frac{G_N M}{r^2} \ .$$

It is similar to the Newton law, but it is not valid for $r > 0$. It can be solved by an ansatz

$$r = r_0\Big(\frac{\tau}{\tau_0}\Big)^{\alpha} \ ,$$

with initial condition $r(\tau_0) = \infty$ and $\dot{r}(\tau) = 0$. α turns out to be $2/3$.

Proof. In fact,

$$\ddot{r} = r_0\tau_0^{-\alpha}\alpha(\alpha - 1)\tau^{\alpha-2} = -\frac{G_N M}{r^2} = -G_N M r_0^{-2}\tau_0^{2\alpha}\tau^{-2\alpha} \ .$$

q.e.d.

Notice that, if we have two radially in-falling particles, starting at the same distance r_0 with the same velocity, they will measure the same proper time. Their arc distance is proportional to $r = r(\tau)$. Therefore, the equation will measure the tidal forces perpendicular to the falling line, since it is proportional to the geodesics. These forces diverges for $r \to 0$. By making a variation of the equation, we find

$$\delta\ddot{r} \simeq \frac{G_N M}{r^3}\delta r \ .$$

This phenomenom is called spaghettification.

The geodesics for massless particle can be found by setting $ds^2 = d\theta^2 = d\phi^2 = 0$. Hence

$$\frac{dr}{dt} = \pm\Big(1 - \frac{2G_N M}{r}\Big) \ .$$

where the plus means that it is falling and minus it is climbing.

6.5 Gravitational red-shift

Consider a radial geodesic for photons, i.e. $T = 0$. Since the metric varies significally from Minkowski only near $r \simeq R_H$, we cannot put an experimental apparatus there. However, we can look at the signal that comes from an in-falling probe, starting from far away.

Since there exists a time-like Killing vector $\underline{K}_t$, its energy is conserved along a geodesic

$$E = -K_t^\mu u_\mu \ ,$$

where u^μ is the 4-velocity of a particle along the geodesic, parametrised by λ.

Proof. In fact

$$\frac{dE}{d\lambda} = -u^\nu \nabla_\nu (K_t^\mu u_\mu) = -K_t^\mu \underbrace{u^\nu \nabla_\nu u_\mu}_{0} + u^\nu u^\mu \nabla_\nu K_{t\mu} = 0 \ ,$$

where we have used $d/d\lambda = u^\nu \nabla_\nu$, the geodesic equation and the Killing equation. q.e.d.

Therefore, the 4-velocity for a static observer U^μ is

$$U^\mu = \frac{K_t^\mu}{\sqrt{-K_t^\nu K_{t\nu}}} \ .$$

Notice that U^μ is normalised at $U^\mu U_\mu = -1$ whereas $\underline{K}_t$ not. The energy measured by this static observer is

$$\omega = -U^\mu u_\mu = -\frac{K_t^\mu u_\mu}{\sqrt{-K_t^\nu K_{t\nu}}} = \frac{E}{|K_t|} \ .$$

Now consider a photon which crosses two different static observer at points r_1 and r_2. They will measure photon's energies

$$\frac{\omega_1}{\omega_2} = \frac{|K_t(r_2)|}{|K_t(r_1)|} \ .$$

In particular, at constant r, we have

$$U^\mu = (\dot{t}(r), 0, 0, 0) = \dot{t} K_t^\mu \ ,$$

and

$$\frac{\omega_1}{\omega_2} = \frac{\dot{t}(r_1)}{\dot{t}(r_2)} = \sqrt{\frac{1 - R_H/r_1}{1 - R_H/r_1}} < 1 \ ,$$

where we assumed $r_1 > r_2$. Physically, it means that the photon seems to lose energy as it climbs. If we add also the relativistic Doeppler effect, due to the velocity v_2, we have the complete formula

$$\omega_1 \simeq \omega_s \sqrt{\frac{1 - v_2}{1 + v_2}} \sqrt{\frac{1 - R_H/r_1}{1 - R_H/r_1}} \ .$$

The gravitational effect depends on the geometry, whereas the Doeppler effect depends on the initial conditions.

Consider a probe sent radially towards the source and a static observer at $r_0 \gg R_H$. The probe emits signals of frequency ω_s at fixed intervals $\Delta\tau_s$, constant since the probe is an inertial frame. The points of emission are $S_n = (\tau_n = n\Delta\tau_s, r_n)$, whereas the point of detection are $O_n = (t_n, r_0)$. The equations of an in-falling probe are

$$\frac{dr_s}{d\tau} = -\sqrt{\frac{R_H}{r_s} - 1 + E^2} \ , \quad \frac{d^2 r_s}{d\tau^2} = -\frac{R_H}{2r_s^2} \ ,$$

where $E^2 = 1 - R_H/r_0 \simeq 1$. Therefore

$$\frac{dr_s}{d\tau} \simeq -sqrt\frac{R_H}{r_s} \ , \quad \frac{dt}{d\tau} = \left(1 - \frac{R_H}{r_s}\right)^{-1} E \simeq \left(1 - \frac{R_H}{r_s}\right)^{-1} \ .$$

The out-going signal is

$$\frac{dr_\gamma}{dt} = 1 - \frac{2G_N M}{r_\gamma} \ , \quad \omega_n = \omega_n' \sqrt{\frac{1 - R_H/r_n}{1 - R_H/r_0}} \simeq \omega_n' \sqrt{1 - \frac{R_H}{r_n}} \ ,$$

where ω_n' is the frequency measured by a static observer in r_n. The interval of detection is composed by two phenomena: in-falling probe $\Delta t_n^{(1)}$ and the difference in travel time $\Delta t_n^{(2)}$. The former is given by

$$\Delta t_n^{(1)} = \frac{\Delta\tau_s}{1 - R_H/r_n} \ .$$

For the latter, the radial difference is

$$\delta r_n = r_n - r_{n-1} = \int_{r_{n-1}}^{r_n} d\tau \frac{dr_s}{d\tau} \simeq -\int_{\tau_{n-1}}^{\tau_n} d\tau \sqrt{\frac{R_H}{r_s(\tau)}} \simeq \sqrt{\frac{R_H}{r_s(\tau)}} \Delta\tau_s \ ,$$

hence

$$\begin{aligned}\Delta t_n^{(2)} = t_n - t_{n-1} &= \int_{r_n}^{r_0} dr_\gamma \frac{dt}{dr_\gamma} - \int_{r_{n-1}}^{r_0} dr_\gamma \frac{dt}{dr_\gamma} = -\int_{r_{n-1}}^{r_n} dr_\gamma \frac{dt}{dr_\gamma} \\ &= -\int_{r_{n-1}}^{r_n} dr_\gamma \Big(1 - \frac{R_H}{r_\gamma}\Big)^{-1} \simeq -\Big(1 - \frac{R_H}{r_\gamma}\Big)^{-1} \Delta r_n \simeq \sqrt{\frac{R_H}{r_n}} \frac{\Delta\tau_s}{1 - R_H/r_n} \ .\end{aligned}$$

Puttin together, we have

$$\Delta t_n = \Delta t_n^{(1)} + \Delta t_n^{(2)} \simeq \Big(1 + \sqrt{\frac{R_H}{r_n}}\Big) \frac{\Delta\tau_s}{1 - R_H/r_n} \ .$$

Physically, it means that the observer will receive less pulse per unit of time.

If the probe remained at fixed radius, we would have only the first term

$$\Delta_n^{(1)} = \frac{\Delta T_s}{\sqrt{1 - R_H/r_n}} \ ,$$

where we have used $\Delta\tau_s = \Delta T_s$. If we interpret $\Delta T_s = 1/\omega_2$, we recover the gravitational red-shift for $r_0 = r_1 \gg R_H$.

Using the total red-shift

$$\omega_n \simeq \omega_s \sqrt{\frac{1 - v_n}{1 + v_n}} \sqrt{1 - \frac{R_H}{r_n}} \ ,$$

we can estimate the velocity with respect to the local static observer

$$v_n = \frac{dr_s}{dT}\Big|_{t_n} = \frac{dr_s}{d\tau}\frac{d\tau}{dt}\frac{dt}{dT}\Big|_{t_n} \simeq \sqrt{\frac{R_H}{r_n}\Big(1 - \frac{R_H}{r_n}\Big)} \ .$$

In particular, for $r_s \to R_H$, we have $v_n \to 0$ and the observed frequency vanishes. Therefore, there is a radius $r_s > R_H$ such that $\omega_n < \omega_c$, i.e. there is a black-out.

6.6 Black holes

The gravitational red-shift implies that a photon would spend all of its energy to escape. Therefore, the Schwarzschild sphere is called the event horizon. No signal can escape from it, in fact for $r > R_H$ there exists both out-going and in-going light conse, for $r \leq R_H$ there exists only in-going (out-going will be stuck on $r = R_H$ or it will contract). The inside is called a black hole. See Table (**??**). It is important to highlight that the real singularity is at $r = 0$ whereas for the event horizons there is only a coordinate singularity. Black hole can be formed by a collapse of a sphere of dust, via neutron stars or white dwarf.

Black hole	Charged	Rotation
Schwarzschild	no	no
Reissner-Nordstroem	yes	no
Kerr	no	yes
Kerr-Newman	yes	yes

Table 6.1: A table of all mathematically known black holes.

6.7 Orbits

Notice that if we multiply by $m/2$, the equation of the energy

$$\dot{r}^2 = \frac{2G_N M}{r} + E^2 - 1 \ ,$$

we find a Newtonian-like expression

$$\frac{m}{2}\dot{r}^2 - \frac{G_N m M}{r} = \frac{m}{2}(E^2 - 1) \ ,$$

where the Newtonian energy is

$$E_N = \frac{m}{2}(E^2 - 1) \ .$$

If $E_N < 0$ or $E^2 < 1$, the orbit is bounded. If $E_N = 0$ or $E^2 = 1$, the orbit is radially in-falling. If $E_N > 0$ or $E^2 > 1$, the orbit is unbounded.

The equation of radial motion of a test particle is

$$\frac{1}{2}\left(\frac{dr}{d\tau}\right)^2 + V(r) = \frac{E^2}{2} = \mathcal{E} \ ,$$

where the potential is

$$V(r) = \frac{\epsilon}{2}\left(1 - \frac{2G_N M}{r}\right) + \frac{L^2}{2r^2} - \gamma\frac{G_N M L^2}{r^3} \ ,$$

with $\gamma = 1$ for general relativity and $\gamma = 0$ for Newtonian gravity. $\epsilon = 1$ for massive particles and $\epsilon = 0$ for massless ones. Now, we delve the solution of this 1-dimensional equation.

Consider circular orbit, whose radius is given by

$$0 = \frac{dV}{dr}\bigg|_{r_c} \sim \epsilon G_N M r_c^2 - L^2 r_c + 3\gamma G_N M L^2 \ .$$

This critical radius is stable if they are a minimum, otherwise it is unstable. In Newtonian gravity for $\gamma = 0$, we have for massless particles no circular orbits. Instead, for massive particles, we have

$$r_c = \frac{L^2}{G_N M} \ .$$

In general relativity $\gamma = 1$, we have for massless particles

$$r_c = 3G_N M \ ,$$

which is unstable. Instead, for massive particles, we have

$$r_\pm = \frac{L^2}{2G_N M}\Big(1 \pm \sqrt{1 - \frac{12G_N^2 M^2}{L^2}}\Big) \ .$$

Hence, for $L > \sqrt{12}G_N M$, the orbit is unstable for r_- and stable for r_+. For large L, r_- recovers the massless case and r_+ recovers the Newtonian case. For small L, they coincide for $L = \sqrt{12}G_N M$ with radius $r_c = 6G_N M$. For $L < \sqrt{12}G_N M$, no circular orbits are possible.

Non-circular orbits can be viewed as ellipses that precess with $r = r(\phi)$. Substituting $L = r^2 d\phi/dr$ for $\theta = \pi/2$, we have

$$\Big(\frac{dr}{d\phi}\Big)^2 = \frac{E^2 - \epsilon}{L^2} r^4 + \frac{2\epsilon G_N M}{L^2} r^3 - r^2 + 2\gamma G_N M r \ ,$$

which becomes for $x = L^2/G_N M r$

$$\Big(\frac{x}{\phi}\Big)^2 = \frac{L^2(E^2 - \epsilon)}{G_N^2 M^2} + 2\epsilon x - x^2 + \gamma \frac{2G_N^2 M^2}{L^2} x^3 \ ,$$

$$\frac{d^2 x}{d\phi^2} = \epsilon - x + \gamma \frac{3G_N^2 M^2}{L^2} x^2 \ .$$

Defining the dimensionless quantities

$$\alpha = \frac{G_N M}{L} \ , \quad \beta = \frac{G_N M}{EL} = \frac{\alpha}{E} \ , \quad \rho = \frac{2G_N^2 M^2}{L^2} = 2\alpha^2 \ ,$$

we obtain

$$\Big(\frac{x}{\phi}\Big)^2 = \Big(\frac{1}{\beta^2} - \frac{\epsilon}{\alpha^2}\Big) + 2\epsilon x - x^2 + \gamma \rho x^3 \ ,$$

$$\frac{d^2 x}{d\phi^2} = \epsilon - x + \frac{3}{2}\gamma \rho x^2 \ ,$$

For Newtonian gravity $\gamma = 0$, we have

$$\Big(\frac{x}{\phi}\Big)^2 = \Big(\frac{1}{\beta^2} - \frac{\epsilon}{\alpha^2}\Big) = -(x - x_+)(x - x_-) \ ,$$

$$\frac{d^2 x}{d\phi^2} = \epsilon - x \ .$$

Calling $x_\mp = \epsilon \mp k$, with

$$k = \sqrt{\frac{1}{\beta^2} - \frac{\epsilon}{\alpha^2} + \epsilon^2} \ .$$

The general solution is

$$x(\phi) = x_- + (x_+ - x_-)\sin^2(\phi/2 + \delta) = \epsilon(x_- - \epsilon)\cos(\phi + 2\delta) \ ,$$

where δ is an integration constant. The eccentricity is given by

$$e = \frac{x_+ - x_-}{x_+ + x_-} = \frac{k}{\epsilon} \ .$$

For massive particles, we have: if $x_- > 0$ and $x_- \neq x_+$ then the orbit is an ellipse with x_- perihelion and x_+ aphelion; if $x_- = x_+$ or $k = 0$ then the orbit is circular; if $x_- = 0$ or $k = 1$ then the orbit is a parabola and x_2 is the vertex; if $x_1 < 0$ or $k > 1$ then the orbit is an hyperbola and x_2 is the vertex. For massless particles, $x \in [-k, k]$ and the orbit is unbounded.

In general relativity $\gamma = 1$, we have

$$\left(\frac{dx}{d\phi}\right)^2 = \rho(x - x_1)(x - x_2)(x - x_3) \ ,$$

with $x_1 + x_2 + x_3 = 1/\rho$. The general solution is

$$x(\phi) = x_1 + (x_2 - x_1)\,\mathrm{sn}^2\left(\frac{\phi}{2}\sqrt{\rho(x_3 - x_1)} + \delta, k\right) \ ,$$

where $sn(z, k)$ is the Jacobi elliptic function with argument z and modulus $k = \sqrt{(x_2 - x_1)/(x_3 - x_1)}$. The roots can be real ($x_1 \leq x_2 \leq x_3$) or one real x_1 and two complex. If $x_1 < x_2 < x_3$ all real then

$$\frac{d^2x}{d\phi^2} = \frac{\rho}{2}\Big((x - x_2)(x - x_3) + (x - x_1)(x - x_3) + (x - x_1)(x - x_2)\Big) \ ,$$

which is positive for $x = x_1$ and $x = x_3$ and negative for $x = x_2$. Physically, we have an oscillation between x_1 and x_2 or move away from x_3 to infinity. If $x_1 < 0$, a particle coming from infinity get near the mass and move again to infinity. If $x_1 < x_2 = x_3$, calling $r_{in} = 1/x_2$, we have $3/2\rho \leq r_{in} \leq r\rho$. Therefore, there are three solutions: spiral approaching the inner radius; circular orbit at the inner radius; spiral down from the inner radius towards the singularity. Fow $k = 1$, we have $\mathrm{sn}(z, 1) = \tanh(z)$ and

$$x(\phi) = x_1 + (x_2 - x_1)\tanh^2\left(\frac{\phi}{2}\sqrt{\rho(x_2 - x_1)} + \delta\right) \ .$$

If $x_1 = x_2 < x_3$ then the orbit is circular in $r_{out} = 1/x_1$. If $x_2, x_3 \in \mathbb{C}$, then the particle spirals towards a black hole with finite ϕ.

Chapter 7

Weak field phenomena

7.1 Perihelion precession

Consider a circular orbit in radius $r_c = L^2/G_N M$. With $\gamma = \epsilon = 1$, we approximate $r = r_c + \delta r(\tau)$. Expanding the potential, we obtain frequency

$$\omega_r^2 = V''(r_c) = \frac{G_N^4 M^4}{L^6} - \frac{6G_N^6 M^6}{L^8} = \omega_N \Big(1 - \frac{6G_N^2 M^2}{L^2}\Big)$$

where ω_N is the Newtonian frequency. Therefore,

$$\delta\phi_T \simeq \frac{2\pi}{\omega_r} - \frac{2\pi}{\omega_N} \simeq \frac{6\pi G_N^2 M^2}{L^2} \simeq \frac{6\pi G_N M}{r_c} \ .$$

Generalising for elliptic orbits, we have

$$\delta\phi_T \simeq \frac{6\pi G_N M}{a(1-e^2)} \ .$$

7.2 Light deflection

Consider a trajectory of light passing near the Sun with $\epsilon = 0$ and

$$\Big(\frac{r}{\phi}\Big)^2 = \frac{E^2}{L^2} r^4 - r^2 + 2\gamma G_N M r \ .$$

In Newtonian gravity, $\gamma = 0$, light travels in straight line with impact parameter $b = L/E$. Therefore

$$\frac{d\phi}{dr} = \pm \frac{1}{r^2}\Big(\frac{1}{b^2} - \frac{1}{r^2}\Big(1 - \frac{\gamma 2 G_N M}{r}\Big)\Big)^{-1/2}$$

which becomes for $u = 1/r$

$$\frac{d\phi}{du} = \left(\frac{1}{b^2} - u^2 + 2\gamma G_N M u^3\right)^{-1/2} .$$

For $\gamma = 0$, the solution is

$$r \sin\phi = b .$$

For $\gamma = 1$ and $G_N M/b \ll 1$, we define $u = u(1 - G_N M u)$ and expand

$$\frac{d\phi}{dy} \simeq \frac{1 + 2G_N M y}{\sqrt{1/b^2 - y^2}} ,$$

which is solved by

$$\phi \simeq \arcsin(by) + \frac{2G_N M}{b} - 2G_N M \sqrt{1/b^2 - y^2} .$$

At the minimum $y = 1/b$, we have

$$\phi_b = \frac{\pi}{2} + \frac{2G_N M}{b}$$

and

$$\delta\phi = 2\phi_b - \pi \simeq \frac{4G_N M}{b} .$$

Chapter 8

Interior Schwarzschild metric

8.1 Metric

Consider a static spherical metric in the form

$$ds^2 = -\exp(\nu)dt^2 + \exp(\lambda)dr^2 + r^2 d\Omega^2 \ .$$

Suppose we are dealing with a perfect fluid with density $\rho(r)$ and pressure $p(r)$. The Einstein equations reads

$$G^0{}_0 = -\exp(-\lambda)\Big(\frac{\lambda'}{r} - \frac{1}{r^2}\Big) + \frac{1}{r^2} = -8\pi G_N \rho \ ,$$

$$G^1{}_1 = \exp(-\lambda)\Big(\frac{\nu'}{r} + \frac{1}{r^2}\Big) + \frac{1}{r^2} = 8\pi G_N p \ ,$$

$$G^2{}_2 = G^3{}_3 = -\exp(-\lambda)\Big(\frac{\lambda''}{2} + \frac{(\nu')^2}{4} - \frac{\nu'\lambda'}{4} + \frac{\nu' - \lambda'}{2r}\Big) = 8\pi G_N p \ ,$$

From the first, we have

$$g^{rr} = \exp(-\lambda) = 1 - \frac{2G_N m(r)}{r} \ ,$$

where the Misner-Sharp-Hernandez mass function is

$$m(r) = 4\pi \int_0^r dx x^2 \rho(x) \ .$$

Physically, it is the total mass star for $r = R_s$. Notice the continuity condition

$$g^{rr}(r \to R_s^-) = g^{rr}(r \to R_s^+) \ .$$

Homogeneity means $\rho(r) = \rho = \rho_0$ for $r \in [0, R_s]$. Therefore

$$m = \frac{4}{3}\pi r^3 \rho_0$$

and

$$\exp(-\lambda) = 1 - \frac{8}{3}\pi G_N \rho r^2 = 1 - \frac{2G_N M r^2}{R_s^3} \ .$$

Solving the remaining, we obtain

$$-g_{tt} = \exp(\nu) = \frac{1}{4}\Big(3\sqrt{1 - \frac{2G_N M}{R_s}} - \sqrt{1 - \frac{2G_N M}{R_s}}\Big)^2 \ .$$

Therefore, the pressure is

$$p = \rho \frac{\sqrt{1 - 2G_N M/R_s} - \sqrt{1 - 2G_N M r^2/R_s^3}}{\sqrt{1 - 2G_N M r^2/R_s^2} - 3\sqrt{1 - 2G_N M/R_s}} \ .$$

Notice that it is again continuous

$$g^{tt}(r \to R_s^-) = g^{tt}(r \to R_s^+) \ .$$

A consequence of the Birkhoff theorem states that pressure must vanish across R_s since it vanishes outside and we use the same coordinates. However, this does not hold for energy, since it is at first order differential.

8.2 TOV equation

The Bianchi identity implies that

$$\nabla_\mu T^\mu{}_1 = p' + \frac{\nu'}{2}(p + \rho) = 0 \ ,$$

which implies that

$$p' = -(\rho + p)\frac{G_N}{r^2}(m + 4\pi r^3 p)\Big(1 - \frac{2G_N m}{r}\Big)^{-1} \ .$$

In the non-relativistic limit it becomes

$$p' \simeq \rho(r)\frac{G_N m}{r^2} = -\rho(r) V_N'(r) \ .$$

As equation of state for a polytropes, we use

$$p = p_0 \Big(\frac{\rho}{\rho_0}\Big)^{1+1/n} \ ,$$

where n is the polytropic index. The pressure is maximal at $r = 0$

$$p(0) = \rho_0 \frac{\sqrt{1 - 2G_N M/R_s} - 1}{1 - 3\sqrt{1 - 2G_N M/R_s}}$$

and diverges for

$$R_S = \frac{9}{4} G_N M = \frac{9}{8} R_H \ .$$

Physically, it means that for $R < R_s$, the star is not in equilibrium.

Part IV

Cosmology

Chapter 9

Friedman-Lemaitre-Robertson-Walker

9.1 Principles

Modern cosmology is based on two principles

Principle 9.1 (Copernican)
There is no preferred observer in the Universe.

Principle 9.2 (Cosmological)
The Universe is homogeneous and isotropic.

Isotropy is taken experimentally and homogeneity follows from isotropy of independent of observation. The average distribution of matter of a sampled area is the same in all directions. There exists a time t such that the Universe is homogeneous and isotropic on each time slice. We will consider that galaxies are an homogeneous fluid.

Using the symmetries, we can define three rotational space-like Killing vectors associated to isotropy and three translational space-like Killing vectors associated to homogeneity. It can be proved that by maximally symmetry, isotropy with respect to an arbitrary point is equivalent to homogeneity. The FWR metric is

$$ds^2 = -dt^2 + a^2(t)\Big(\frac{dr^2}{1-kr^2} + r^2 d\Omega^2\Big) = -dt^2 + a^2(t)d\sigma^2 \ .$$

The coordinates are (t, r, θ, ϕ). a is the scale factor. k could take any value but with a suitable rescaling of r

$$dr = \sqrt{|k|}dr \ , \quad a^2 = |k|a^2 \ ,$$

k could be $0, \pm 1$, which means: if $k = 0$, we have a flat universe

$$d\sigma^2 = dr^2 + r^2 d\Omega^2 = dx^2 + dy^2 + dz^2 \ ;$$

if $k = 1$, we have a closed universe for $r = \sin x$

$$d\sigma^2 = dx^2 + \sin^2 x d\Omega^2 \ ;$$

if $k = -1$, we have a closed universe for $r = \sinh x$

$$d\sigma^2 = dx^2 + \sinh^2 x d\Omega^2 \ .$$

The areal radius is given by

$$r_A = a(t) r \ ,$$

and the proper distance is given by

$$dR = a(t) \frac{dr}{\sqrt{1 - kr^2}} \ .$$

Observations suggest that local length does not change whereas distances increases as the scale factor.

9.2 Cosmic fluid

We assumed that the Universe is a perfect fluid. The 0-th components of the energy momentum tensor is

$$-\nabla_\mu T^\mu{}_0 = \dot{\rho} + 3H(\rho + p) = 0 \ ,$$

where the Hubble constant $H = \dot{a}/a$. Therefore,

$$\frac{d}{dt}(a^3 \rho) = -p \frac{d}{dt}(a^3) \ .$$

Assume that $p = \omega \rho$, then

$$\frac{\dot{\rho}}{\rho} = -3(1 + \omega) \frac{\dot{a}}{a}$$

and

$$\rho \propto a^{-3(1+\omega)} \ .$$

The cosmic fluid can be decomposed in dust, i.e. non-relativistic particles, and radiation, i.e. highly-relativistic particles. For the dust case, we have $\omega = 0$ and $\rho \propto a^{-3}$. Notice that it agrees with $E \sim m_0$ and $V \sim a^3$. Constant lines of (r, θ, ϕ) are geodesics. For the radiation case, we have $T = -\rho + 3p = 0$, therefore $\omega = 1/3$. Then $\rho \propto a^{-4}$. Notice that it agrees with the gravitational red-shift $E \propto a^{-1}$. Furthermore, there is another possible source, compatible with the acceleration of the Universe. We have $\omega = -1$. It is called cosmological constant Λ such that

$$\rho = -p = \frac{\Lambda}{8\pi G_N} \ .$$

9.3 Friedmann equations

Starting from the Einstein equations, we find the two Friedmann equations

$$3\left(\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2}\right) = 8\pi G_N \rho \ ,$$

$$3\frac{\ddot{a}}{a} = -4\pi G_N(\rho + 3p) \ .$$

The first one is a constraint which selects the initial conditions. Its time derivative is

$$6\frac{\dot{a}}{a}\left(\frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2} - \frac{k}{a^2}\right) = 8\pi G_N \dot{\rho} \ .$$

Using the first equation, we find

$$3\frac{\ddot{a}}{a} - 8\pi G_N \rho = -12\pi G_N(p + \rho) \ .$$

Now, we define the deceleration parameter $q = -a\dot{a}/\dot{a}^2$ and the density parameter $\Omega = 8\pi G_N \rho/3H^2 = \rho/\rho_{critical}$, where $\rho_{critical} = 3H^2/8\pi G_N$. The Friedmann equations becomes

$$\Omega - 1 = \frac{k}{H^2 a^2} \ ,$$

which implies that

1. open universe for $\rho < \rho_{critical}$ for $k = -1$ or $\Omega < 1$,
2. flat universe for $\rho = \rho_{critical}$ for $k = 0$ or $\Omega = 1$,
3. closed universe for $\rho > \rho_{critical}$ for $k = 1$ or $\Omega > 1$.

Observations suggest that $k = 0$. For a matter dominated universe, we have

$$\rho_{dust} \sim \frac{1}{a^3} \ , \quad \frac{\dot{a}^2}{a^2} \sim \frac{1}{a^3} \ ,$$

$$a\dot{a}^2 \sim 1 \ , \quad \sqrt{a}da \sim dt \ ,$$

therefore

$$a^{3/2} \sim t \ .$$

For a radiation dominated universe, we have

$$\rho_{rad} \sim \frac{1}{a^4} \ , \quad \frac{\dot{a}^2}{a^2} \sim \frac{1}{a^4} \ ,$$

$$a^2\dot{a}^2 \sim 1 \ , \quad ada \sim dt \ ,$$

therefore

$$a^2 \sim t \ .$$

For a matter dominated universe, we have

$$\rho_\Lambda \sim \Lambda \ , \quad \frac{\dot{a}^2}{a^2} \sim \frac{\Lambda}{3} \ ,$$

$$\sqrt{\frac{\Lambda}{3}} \sim \frac{\dot{a}}{a} = H_0 \ ,$$

therefore

$$a \sim \exp(H_0 t) \ ,$$

where H_0 is the true cosmological constant.

Consider a particle of mass m on the surface of radius $R = ra$. The Newtonian acceleration

$$\ddot{R} = r\ddot{a} = -\frac{G_N 4\pi\rho R^3}{3R^2} = -\frac{4}{3}\pi G_N \rho r a \ , \quad 3\frac{\ddot{a}}{a} = -4\pi G_N \rho \ .$$

Notice that it is different from Einstein, since it is true only if $p = 0$.

Chapter 10

Expansion of the universe

10.1 Cosmological red-shift

Consider a source r_1 and a receiver r_2 moving along a cosmic fluid. Two emitted signals at times t_1 and $t_1 + \delta t_1$ will be received at t_2 and $t_2 + \delta t_2$, after

$$\int_{t_1}^{t_2} \frac{dt}{a(t)} = \int_{r_1}^{r_2} \frac{dr}{\sqrt{1-kr^2}} = \int_{t_1+\delta t_1}^{t_2+\delta t_2} \frac{dt}{a(t)} \ ,$$

which implies that

$$\int_{t_1}^{t_1+\delta t_1} \frac{dt}{a(t)} = \int_{t_2}^{t_2 \delta t_2} \frac{dt}{a(t)} \ .$$

Assuming δt_1 and δt_2 are small, then

$$\frac{\delta t_1}{a(t_1)} = \frac{\delta t_2}{a(t_2)} \ .$$

Therefore,

$$\frac{\omega(t_1)}{\omega(t_2)} = \frac{a(t_1)}{a(t_2)} = \frac{a_1}{a_2} \ .$$

Introducing the red-shift factor

$$z = \frac{\lambda_0 - \lambda_s}{\lambda_s} = \frac{a_0}{a_s} - 1 \ ,$$

where λ_0 is observed and λ_s is emitted. The red-shift is caused by the expansion of the Universe. It is important notice that a can be rescaled and therefore is not physical, however its ratio does.

10.2 Luminosity-distance relation

Consider the flux of energy measured by an observer in Minkowski. Since energy is conserved across a sphere, we can define the intrinsic luminosity $L_0 = E_0/T_0$. Therefore, the luminosity-distance relation is

$$F = \frac{L}{A} = \frac{L_0}{4\pi R^2} \ , \quad d_L^2 = R^2 = \frac{L_0}{4\pi F} \ .$$

In the FRW metric, photons are red-shifted by $(1+z)$. Moreover, the frequency is reduced between observer and emitter by a factor δt. Therefore,

$$L \simeq \frac{L_0}{(1+z)^2} \ , \quad F = \frac{L}{A} \simeq \frac{L_0}{4\pi (a_0 r)^2 (1+z)^2} = \frac{L_0}{4pid_L^2} \ ,$$

and

$$d_L = a_0 r(1+z) \simeq \sqrt{\frac{L_0}{F}} \ . \tag{10.1}$$

In order to measure the red-shift, we need the emitted frequency. This can be found by observing the period of oscillation of the luminosity and the absolute luminosity. These sources are called standard candle and form cosmic distance ladder.

10.3 Hubble-Lemaitre law

A photon is gravitationally red-shifted by

$$z = \frac{a_0}{a_s} - 1 \ ,$$

therefore, along a light ray for $kr^2 \ll 1$, we find

$$ds^2 = -dt^2 + \frac{a^2 dr^2}{1-kr^2} = 0$$

and

$$\int_{t_s}^{t_0} \frac{dt}{a(t)} = \int_0^r \frac{dr}{\sqrt{1-kr^2}} \simeq \int_0^r dr = r \ .$$

Expanding the cosmic factor

$$a(t) = a_0 + \dot{a}_0 (t-t_0) + \frac{\ddot{a}_0}{2}(t-t_0)^2$$

then

$$r = \frac{1}{a_0}\Big((t_0 - t) + + \frac{H_0}{2}(t_0 - t)^2\Big)$$

and

$$\frac{1}{1+z} = \frac{a_s}{a_0} = 1 + H_0(t-t_0) - \frac{1}{2}q_0 H_0^2 (t-t_0)^2$$

where

$$q_0 = -\frac{a_0 \ddot{a}_0}{\dot{a}_0^2} = \frac{1+3\omega}{2}\Omega \ .$$

For small $H_0(t-t_0)$, we have

$$t_0 - t = \frac{1}{H_0}\Big(z - \Big(1 + \frac{q_0}{2}z^2\Big)\Big) \ .$$

Finally, the Hubble law is

$$d_L = \frac{1}{H_0}\Big(z + \frac{1}{2}(1-q_0)z^2\Big) \simeq \frac{z}{H_0} \ .$$

Physically, it means that galaxies recede from us wit a velocity z.

Chapter 11

Standard model of cosmology

11.1 CMB and inflation

The solutions of the Friedmann equations shows that the Universe must have started from a singularity at $a = 0$ and expands from that. This is the basis of the Hot Big Bang model. Its main adversary is the steady model, in which energy density remains constant. However, the Cosmic Microwave Background radiation shows that the Universe is cooling down.

The CMB radiation is an almost relic radiation, generated by the decoupled of radiation from matter on the last scattering surface. Before, photons were in thermal equilibrium with electrons and protons. They only way to understand the physics of that period is through gravitational waves.

The light we receive from different direction in the sky should not have the same temperature, because they were not causally connected. In fact

$$ds^2 = 0 \ , \quad dt = adr \ , \quad dr = \frac{dt}{a} \ .$$

Suppose we are at $r = 0$ and find

$$r_s = \int_0^{r_s} dr \sim \int_{-t_s}^0 \frac{dt}{a(t)} \ .$$

For a matter or radiation dominated Universe, we have

$$a(t) \sim t^\alpha \ , \quad r_s \sim t_s^{1-\alpha} \ ,$$

where $\alpha \in (0, 1)$. The proper distance is

$$R \sim a(-t_s) r_s = t_s^\alpha t_s^{1-\alpha} = t_s \ ,$$

which coincides with Minkowski. Furthermore,

$$\dot{a} \sim t^{\alpha-1} \ , \quad H = \frac{\dot{a}}{a} \sim t^{-1} \ ,$$

and the particle horizon grows with time

$$R_H \sim \frac{1}{H(-t_s)} \sim t_s \ . \tag{11.1}$$

However, in a vacuum dominated Universe

$$a(t) \sim \exp(\exp(H_0 t)) \ , \quad r_s \sim \frac{\exp(-H_0 t)}{H_0} \ ,$$

and the particle horizon is constant

$$R \sim \exp(H_o t_s) r_s \sim \frac{1}{H_0} \ .$$

This can explain CMB homogeneity: the Universe started so small to have all particles in causal connection. Then, there was an inflation phase of rapid expansion. Finally, it ended.

11.2 ΛCDM model

Observations lead to a flat Universe with $\Omega \simeq 1$ with density $\rho_0 = \rho_{crit} \simeq 10^{-29} g/cm^3$, identified by three sources

1. massive baryonic matter with $\rho_{mat}/\rho_0 \simeq 5\%$
2. dark matter with $\rho_{DM}/\rho_0 \simeq 25\%$
3. dark energy with $\rho_{DE}/\rho_0 \simeq 70\%$

Dark matter is needed to explain the rotation of galaxies.

Bibliography

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